

GEORGIA DEPARTMENT OF TRANSPORTATION

Resilience Improvement Plan

March 2025



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Executive Summary

The Georgia Department of Transportation (GDOT) is committed to building and maintaining transportation infrastructure designed to withstand and/or rapidly recover from the impacts associated with hazard or extreme events.

This Resilience Improvement Plan (RIP) establishes GDOT's resilience goals; outlines existing resilience best practices, efforts, laws, and tools; assesses risk and vulnerability of the state's transportation network; recommends resilience improvement policies and strategies; sets targets; and includes a prioritized list of projects eligible for Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Formula Program funds.

The plan was developed within the framework of the GDOT 2050 Statewide Transportation Plan /Statewide Strategic Transportation Plan (SWTP/SSTP) and aligned with the Transportation Asset Management Plan. This framework is as follows:

FOUNDATIONAL



- Reduce risk to transportation infrastructure, operations, disadvantaged populations, and aged individuals.
- Determine which hazards pose the highest risk and the areas of those risks
- Prioritize projects within high-risk/high hazard areas

CATALYTIC



- Ensure today's investments meet tomorrow's growth
- Promote investments minimizing or reducing operational and response costs
- Ensure investments are not creating new risks

INNOVATION



- Collaboration with federal and state partners to address and reduce transportation network risk
- Build on vulnerability assessment and the Multi-Hazards Exposure and Vulnerability Analysis Tool (MHEVRA) to identify existing and future risks
- Promote investments in natural infrastructure to minimize future risks

Within the framework of these foundational, catalytic, and innovation goals, the state of Georgia's infrastructure leads the nation. Ranked #1 in the country and graded A+ for the best infrastructure by CNBC¹, 98% of the bridges and 94% of the roadways in the state are in acceptable condition. The RIP will add an enhanced resilience lens in maintaining the state's transportation assets.

The plan development process included establishment of a Resiliency Committee, a peer review, and a detailed review of existing resilience policies, laws, and tools. A focal point of the RIP is a Climate Vulnerability, Risk, and Resilience Assessment of Georgia's pavements, bridges, and culverts developed by Georgia Tech for GDOT as a research study.

The RIP recommends strategies to increase the resilience of the transportation system. The strategies address the planning process through transportation system management and operations.

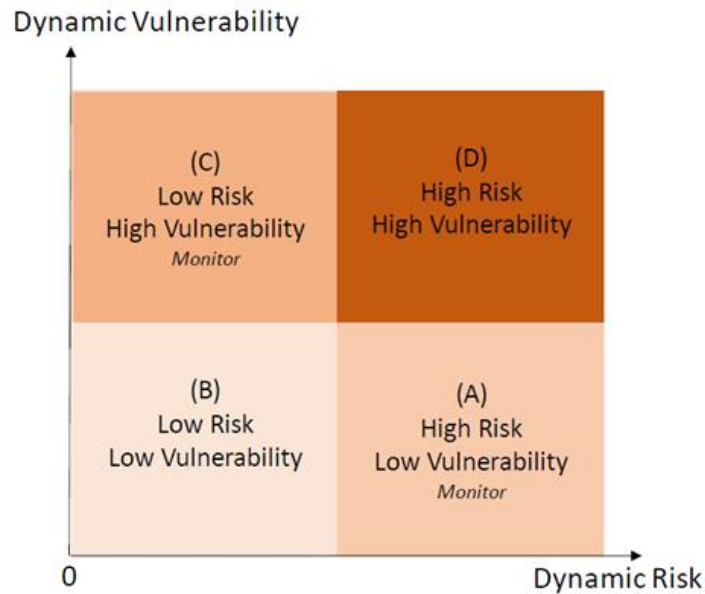
¹ <https://www.cnbc.com/2023/07/18/these-are-americas-best-states-for-infrastructure.html>

The focus of the RIP is identifying vulnerabilities through a risk-based assessment and how the plan improves the ability to respond to events. Performance metrics were developed based on the plan goals and findings from the Assessment. These metrics will be used to track progress towards continuing to make the transportation system more resilient over time. The metrics focus on the criticality of the infrastructure assets, which is the importance or value of the asset, in terms of the cost to users, owners, and society from a loss in function. A critical asset is an asset that is so important to a study area that its removal would result in significant losses. The metrics also focus on the vulnerability of the assets, which is the degree to which a system is susceptible to, or unable to handle the adverse effects of extreme weather events

These performance measures are aligned with the foundational, catalytic, and innovation goals identified for ensuring a resilient transportation system. With the identification of the high hazard areas and infrastructure ranked with high vulnerability and high criticality, the risk to the mobility of the state's residents is reduced; and identifies opportunities for investments on those areas with the greatest impact on improving resiliency.

A detailed screening based on the project eligibility criteria in the PROTECT legislation of all projects included in Statewide Transportation Improvement Program (STIP) was conducted utilizing the approved FHWA form found in Appendix B. In addition, the projects included in the Transportation Improvement Programs (TIP)/Metropolitan Transportation Plans (MTP) for each Metropolitan Planning Organization (MPO) in the state were also screened utilizing the form. The next step in the process is for the projects determined to be eligible through the screening to be reviewed by both the GDOT Resiliency Committee and FHWA for concurrence. However, since this plan is dynamic and projects will be continuously reviewed and evaluated and new eligible projects identified, it is likely that the projects eligible for PROTECT funding will exceed available funding and that a system for determining the allocation of PROTECT funding to projects will be needed.

As a result, GDOT developed an initial methodology to score projects based on each asset's dynamic risk (product of criticality and dynamic vulnerability) determined using an Asset Prioritization Matrix developed by Georgia Tech.



Source: Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management Draft Final Report, GA Tech

The RIP and associated eligible and prioritized project lists are dynamic and will be updated as needed to ensure features such as new data and data sources, new projects, changes in priorities, and new guidance are incorporated into future iterations.

Additionally, GDOT will review the updated STIP to see if there have been any changes to projects determined eligible for STIP funding and if those changes affect PROTECT funding eligibility. GDOT will also identify new projects that need to be screened, reviewed by the GDOT Resiliency Committee for a recommendation, and sent to FHWA for an eligibility determination. GDOT will update its project list accordingly.

Definitions

Adaptive Capacity: Ability of a system to adjust, repair, and respond to damage or disruption.

Criticality: Importance or value of infrastructure asset, in terms of the cost to users, owners, and society from a loss in function. A critical asset is an asset that is so important to a study area that its removal would result in significant losses.²

Disadvantaged Populations: Populations who belong to disadvantaged communities that are affected by high poverty, low wealth, low educational attainment, insufficient local jobs, high inequality, and low homeownership. Typically measured at the census tract level, these populations can also include people who are historically burdened by lack of access to transportation, health care, environmental quality, and economic opportunities, among others.³

Discretionary Funding: Money allocated through annual appropriations bills and typically assigned to projects via competitive processes such as grant programs. PROTECT discretionary funding is highly competitive and available to entities and projects based on eligibility.

Executive Order: An official directive from the U.S. president to federal agencies.

Exposure: Whether an asset or system is in an area experiencing direct effects of extreme natural, technological, or human-caused hazards.

Formula Funding: Federal funds distributed to states by a statutory formula.

Hazard: A source of potential danger or adverse condition.

Infrastructure Investment and Jobs Act (IIJA)/Bipartisan Infrastructure Law (BIL): A law that reauthorizes surface transportation projects for five years and provides \$550 billion over fiscal years 2022 through 2026 in new federal funds for infrastructure, with half of that funding going to transportation.⁴

Metropolitan Transportation Plan (MTP): A comprehensive blueprint for area transportation services aimed at meeting mobility needs through the next 20+ years.

National Highway Performance Program (NHPP): A program that provides support for the condition and performance of the, for the construction of new facilities on the NHS, and to ensure that investments of Federal-aid funds in highway construction are directed to support progress toward the achievement of performance targets established in a state's asset management plan for the NHS.

National Highway System (NHS): Roadways important to the nation's economy, defense, and mobility.⁵

² Georgia Tech research is based on NCHRP Research Report 970.

(<https://nap.nationalacademies.org/catalog/26125/mainstreaming-system-resilience-concepts-into-transportation-agencies-a-guide>)

³ [SS4A Underserved Communities Census Tracts \(Historically Disadvantaged Communities\) \(arcgis.com\)](https://arcgis.com)

⁴ <https://www.phmsa.dot.gov/legislative-mandates/bipartisan-infrastructure-law-bil-infrastructure-investment-and-jobs-act-iiija>

⁵ https://www.fhwa.dot.gov/planning/national_highway_system/

Natural Hazard: A natural phenomenon that can produce damaging disruptions to systems and their functionality, such as hurricanes, flooding, and wind.

Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Program: A program, with both formula and discretionary funding sources, initiated through the BIL to make transportation infrastructure more resilient to weather events and other natural disasters by focusing on resilience planning, making resilience improvements to existing transportation assets and evacuation routes, and addressing at-risk highway infrastructure.

Resilience: The ability to anticipate, prepare for, adapt to changing conditions and withstand, respond, and recover rapidly from disruptions.

Risk: A combination of the likelihood that an asset will experience an extreme event's impact, and the severity or consequence of that impact. Risk can also be defined as the potential for harm expressed as the product of the probability and the consequence of an event and often expressed in relative term as high, moderate, and low.⁶

Sensitivity: Whether the asset may be damaged or disrupted by a stressor. The more sensitive an asset, the more likely it is to be damaged or disrupted by a stressor.

State Long Range Transportation Plan (SLRTP): A long-range plan that assesses a state's multimodal transportation system and presents plans for improvement and maintenance.

Strategic Highway Network (STRAHNET): The network of highways, including the Interstate System that provides connecting routes to military installations, industries, and resources. It is a subset of the NHS and important to the United States' strategic defense policy by providing defense access, continuity and emergency capabilities for defense purposes.

Statewide Strategic Transportation Plan (SSTP): The SSTP is a state mandated plan that includes the overall transportation investment strategies focused on advancing economic growth in the state.

Statewide Transportation Plan (SWTP): The SWTP is a federally required long-range plan that assesses the state's multimodal transportation system and presents plans for improvements and maintenance through a minimum horizon of twenty years.

Statewide Transportation Improvement Program (STIP): The STIP is the state's four-year transportation program that includes all federally funded transportation projects located outside of the Metropolitan Planning Organization (MPO) boundaries. Each MPO develops their own Transportation Improvement Program (TIP), which are included in the STIP by reference.

Transportation Asset Management (TAM): A strategic and systematic process of operating, maintaining, upgrading, and expanding physical assets effectively throughout their life cycle. It focuses on business and engineering practices for resource allocation and utilization, with the objective of better decision making based upon quality information and well-defined objectives.

Transportation Systems Management and Operations (TSMO): Integrated strategies to optimize the performance of existing infrastructure through the implementation of multimodal and

¹GDOT 2050 STP/2020 SSTP Resilience Analysis

intermodal, cross-jurisdictional systems, services, and projects designed to preserve capacity and improve security, safety, and reliability of the transportation system.

Vulnerability: The degree to which a system is susceptible to, or unable to cope with adverse effects of weather events. In the transportation context, weather vulnerability is a function of the transportation system's exposure to weather effects, sensitivity to those weather effects, and its adaptive capacity.

Introduction

This Resilience Improvement Plan (RIP) identifies immediate and long-range planning activities and investments of the state with respect to the resilience of the surface transportation system through 2050. The plan identifies a systematic approach to transportation system resilience. It includes a risk-based assessment of vulnerabilities of transportation assets and systems to weather events and natural disasters, the ability to respond to these events, resilience improvement policies, and an investment plan that includes a list of priority projects.

Purpose

The purpose of the RIP is to improve the ability of the state to prepare for and respond to weather events, natural disasters, and changing conditions by identifying risks and vulnerabilities, developing proposed resilience solutions, and prioritizing resilience improvements.

The development of the RIP arose from the Infrastructure Investment and Jobs Act/Bipartisan Infrastructure Law (IIJA/BIL) Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Program. The program is designed to help make surface transportation more resilient to the impacts of natural hazards through support of planning activities, resilience improvements, and evacuation routes, and at-risk coastal infrastructure.

Plan Goals

The RIP goals were developed within the framework of the GDOT 2050 SWTP/SSTP which includes three main categories: Foundational, Catalytic, and Innovation (see **Figure 1**).

Figure 1: GDOT RIP Goals

FOUNDATIONAL	
	• Reduce risk to transportation infrastructure, operations, disadvantaged populations, and aged individuals. • Determine which hazards pose the highest risk and the areas of those risks* • Prioritize projects within high-risk/high hazard areas*
CATALYTIC	
	• Ensure today's investments meet tomorrow's growth* • Promote investments minimizing or reducing operational and response costs • Ensure investments are not creating new risks
INNOVATION	
	• Collaboration with federal and state partners to address and reduce transportation network risk* • Build on vulnerability assessment and the Multi-Hazards Exposure and Vulnerability Analysis Tool (MHEVRA) to identify existing and future risks • Promote investments in natural infrastructure to minimize future risks

* Denotes short term goals

Background

What is Resilience?

The PROTECT Formula Program Implementation Guidance defines resilience as the “ability to anticipate, prepare for, or adapt to conditions or withstand, respond to, or recover rapidly from disruptions, including the ability to 1) resist hazards or withstand impacts from weather events and natural disasters; or 2) reduce the magnitude or duration of impacts of a disruptive weather event or natural disaster on a project, and 3) have the absorptive capacity, adaptive capacity, and recoverability to decrease project vulnerability to weather events or other natural disasters.”⁷

Resilience within the transportation system is the ability of the system to respond and maintain operation before, during, and after hazardous events. Though many hazards or stressors require the closure of transportation services, the ability of these services to quickly re-open at or near capacity is indicative of a resilient system.

Resilience strategies are typically grouped into the following four areas⁸:

- Defend
 - Implementation of hard, protective measures
- Accommodate
 - Alteration or adaptation of existing infrastructure
- Retreat
 - Relocation of existing infrastructure
- Make Changes in Policy and Practice
 - Policy modifications that reduce exposure to risks

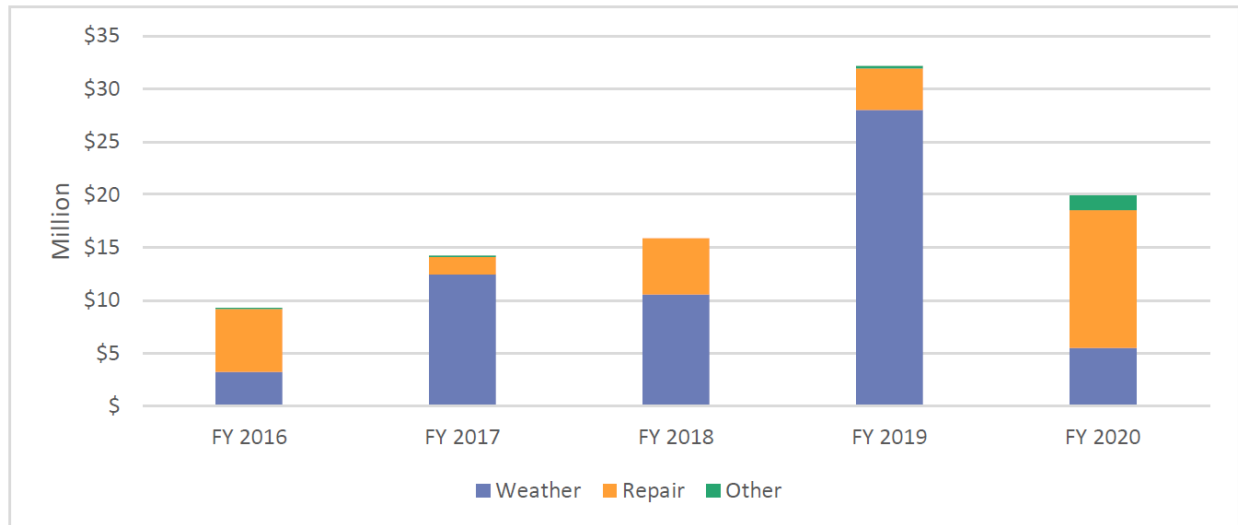
A resilient transportation system is critical to both individuals and communities. While many hazards are unpredictable and beyond human control, planning and foresight can allow the transportation system to continue to function during and after an event. A more resilient transportation system leads to reduced losses and damages, fewer disruptions, and quicker recovery after a disruption.

Per a GDOT 2020 Resiliency Analysis, GDOT spent over \$91 million dollars on emergency projects from 2016 to 2020. Over 65 percent of these funds were used in responding to extreme weather-related events, about 33 percent was used to repair infrastructure damage, and about one percent was used for other emergencies (including COVID-19) (see **Figure 2**).

⁷ https://www.fhwa.dot.gov/environment/sustainability/resilience/policy_and_guidance/protect_formula.pdf

⁸ Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management Draft Final Report

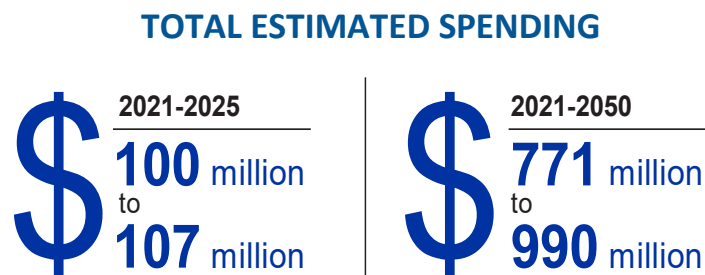
Figure 2: GDOT Emergency Project Spending (FY 2016- FY 2020)



Source: GDOT 2050 SWTP/2020 SSTP Resiliency Analysis

Using the historic emergency events and spending and the projected weather trends for the state, GDOT's emergency spending is estimated to increase almost 30% by 2050, and total spending between 2021 and 2050 is projected to be between \$771 million and \$990 million. (see **Figure 3**).

Figure 3: Total Estimated Emergency Spending



Source: GDOT 2050 SWTP/2020 SSTP Resiliency Analysis

National Policy Context

FHWA has responded to the growing number of hazards facing state and regional transportation agencies which threaten the federal investment in transportation infrastructure by elevating resilience to a national priority. The focus on resilience and sustainability have come through a series of Executive Orders, policies, final rulemaking, and legislation.

Beginning in 2009, Executive Order 13614 established an “integrated strategy towards sustainability” for the federal government and directed all agencies to develop sustainability targets and strategic sustainability plans.

Building on these actions, subsequent Executive Orders focused on improving resilience through engaging in partnerships and information sharing among agencies, risk-based decision making, preparedness planning, and the establishment of a Federal Flood Risk Management Standard. The FHWA also issued an order including a provision of technical assistance to state DOTs and MPOs in undertaking vulnerability assessments and the development of strategies to offset impacts to the transportation system.

The final rule for the development of State Asset Management Plans described the process for developing State risk-based asset management plans. The plans are intended to document how the State DOT will make risk-based decisions as it relates to managing its physical assets and laying out a set of investment strategies to address the condition and system performance gaps.

In addition to the final rulemaking, the White House Council on Environmental Quality also provided the final guidance for Federal agencies on the consideration of the impacts of their actions pertaining to the National Environmental Policy Act reviews, allowing decision-makers and the public to understand the potential impacts of all proposed Federal actions.

The most recent federal transportation legislation has also included resilience initiatives. The Fixing America’s Surface Transportation Act (FAST Act) expanded the scope of the metropolitan planning process aimed at improving resiliency and reliability while reducing vulnerabilities to transportation infrastructure. For the statewide transportation planning process, the FAST Act included the consideration of projects, strategies, and services to improve the resiliency and reliability of the transportation system.

The Infrastructure Investment and Jobs Act (IIJA) or Bipartisan Infrastructure Law (BIL) was passed in 2021 and allows federal funding to be used for “protective features” associated with risk mitigation. The BIL also established new programs through formula funding and discretionary grants to support increasing the resilience of the surface transportation system.

RIP Development

GDOT took several steps prior to developing the RIP in order to create a comprehensive overview of resilience to provide context for the new plan. These steps included forming a Resiliency Committee; conducting a peer review; and documenting existing resilience plans and studies, efforts and policies, laws, and tools.

Resiliency Committee

A Resiliency Committee was formed prior to the plan development process. The Resiliency Committee is responsible for providing data, developing plan goals, submitting projects for

consideration⁹, establishing the screening criteria, approving the prioritization methodology, and reviewing the plan document, as well as reviewing and approving project recommending eligibility for PROTECT formula funding.

The Resiliency Committee is comprised of representatives from the following GDOT offices.

- Office of Planning
- Office of Design Policy and Support
- Office of Maintenance
- Office of Traffic Operations
- Office of Financial Management
- Office of Program Delivery
- Office of Environmental Services
- Office of Roadway Design
- Office of Bridge Design and Maintenance

An FHWA environmental representative also serves as a member of the Resiliency Committee.

Coordination with MPOs

The RIP recognizes the efforts from the MPOs in planning for resilience improvement under the Existing Georgia Plans and Studies section. Many projects included in the current RIP were from the MPO's Transportation Improvement Plans (TIPs), mandated planning documents developed through the ongoing coordination between the MPOs and GDOT, via a screening process for PROTECT Formula funding. In the Next Steps section in the RIP, GDOT also commits to update the data for screening and revisiting the project lists from the updated MTPs, TIPs, and STIP.

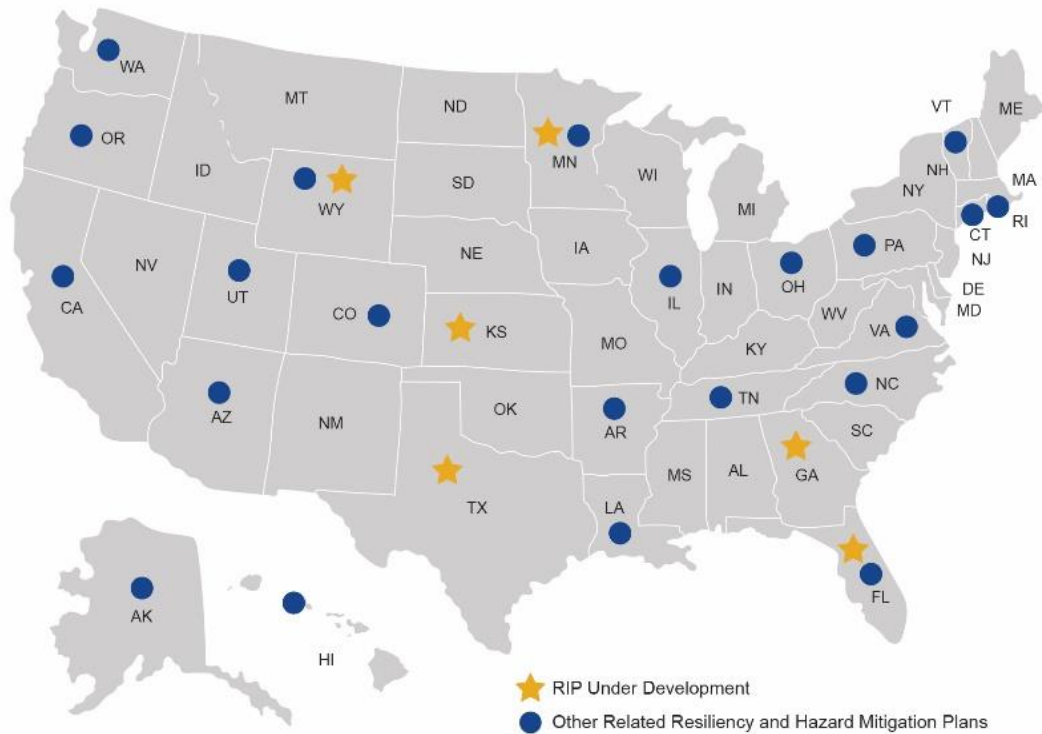
GDOT will continue to coordinate with MPOs to update the data for project screening and selection criteria, and revisit the project lists from the updated MTPs, TIPs, and STIP.

Peer Review and Best Practices

An extensive peer review was conducted to identify best practices related to resiliency and hazard assessments. **Figure 4** depicts the states where plans were reviewed. The list of plans reviewed, and additional findings are available in Appendix A. At the time the peer review was conducted, no other state DOT had completed a RIP; however, a significant number of peers had created resiliency and hazard mitigation plans related to the transportation network and several RIPs were under development. As such, a review of these similar planning efforts was conducted to inform the GDOT RIP development and compare with previous and ongoing resiliency and mitigation efforts in Georgia. These efforts can be divided into one of two categories: Hazard Identification and Assessment and Resilience or Mitigation Strategies.

⁹ The Resiliency Committee recognized established selection processes for TMA, non-TMA, and non-MPO project selection. The process the Resiliency Committee used for submitting projects for consideration outside of these established processes were consistent with these procedures.

Figure 4: Peer Review Inventory



Best Practices Related to Hazard Identification and Assessment

Several peer agencies have developed Hazard Identification and Assessment documents, with a specific focus on the trends within their state. Hazards vary substantially from volcanoes in Hawaii, hurricanes in Florida to wildfires in California. Though varied, the majority of natural, technological, and man-made hazards can be applicable across the nation with higher probability and risk in some areas than others. The hazard identification and assessment best practices documented in these plans include:

- Identify areas that have experienced/are most likely to experience a high number of hazard events
- Identify the hazards that pose the highest risk
- Identify high exposure areas
- Identify frequent/repeated loss infrastructure
- Identify climate trends
- Identify critical infrastructure, utilities, and other assets
- Consider adaptive capacity of the network

Best Practices Related to Resilience and Mitigation Strategies

Similarly, peer state agencies have developed resilience and mitigation plans that highlight areas for improvement. Typically, these plans incorporate hazard assessment results, but don't identify

specific project level improvements. Recommendations such as action plans and the incorporation of resilient strategies were common. The best practices related to resilience and mitigation strategies are as follows:

- Funding and Monetary Considerations
 - Include provisions for projects intended to benefit disadvantaged populations
 - Highlight projects/efforts that can be streamlined for development
 - Differentiate between formula and discretionary funding sources
 - Identify efforts more appropriate for each
- Project and Effort Prioritization
 - Develop a tool and/or matrix to prioritize efforts
 - Identify the frequency for project and effort list updates
- Consider resiliency when designing and constructing transportation projects
- Partner with peer agencies and coordinate with disadvantaged populations
- Develop a plan vision, goals, objectives, and/or policies
- Develop performance measures

Existing Georgia Plans and Studies

A review of existing Georgia plans and studies was conducted to understand how resilience is incorporated into state plans and activities. This review increased alignment with existing plans and studies across the state.

2050 Statewide Transportation Plan (SWTP)/2021 Statewide Strategic Transportation Plan (SSTP)

The GDOT SWTP is a federally required long-range plan that assesses the state's multimodal transportation system and presents plans for improvements and maintenance through a horizon year of at least twenty years, with the current horizon year of 2050. The SSTP is the state mandated plan that identifies the strategic transportation investments focused on supporting economic growth within the state.

The plan is organized into three investment categories (or focus areas) and three investment strategies and identifies future strategies (or action items) to advance planning and collaboration in three areas for each category. **Table 1** shows the action items in the plan associated with resiliency and the focus areas to which the action items relate.

Table 1: 2050 SWTP/2021 SSTP Investment Categories

Action Item	2050 SWTP/2021 SSTP Focus Area
Finalize multi-agency assessment regarding Georgia's level of risk and resiliency	Statewide Freight and Logistics; People Mobility in Metro Atlanta; People Mobility in Emerging Metros and Rural Georgia
Explore R&D and pilot tests for freight technologies (commercial vehicle/ rail safety, first/last mile, platooning, etc.) and supply chain resilience	Statewide Freight and Logistics
Develop policy framework and performance measures for freight fluidity and supply chain resilience	Statewide Freight and Logistics

The 2050 SWTP/2021 SSTP also includes GDOT's first comprehensive resiliency analysis which assesses GDOT's spending in response to weather-related events, the level of risk to the transportation network in Georgia, and resiliency of the system to potential natural, technological, and human-made hazards.

The RIP supports and builds upon the 2050 SWTP/2021 SSTP with goals that are within the framework of the SWTP/SSTP and by incorporating data from the resiliency analysis into the RIP and the Climate Vulnerability, Risk, and Resilience Assessment.

State Mitigation Plan

The PROTECT Formula Program requires the RIP to be consistent with and complementary to the State Mitigation Plan. Georgia's Hazard Mitigation Plan was most recently updated in 2019 by the Georgia Emergency Management and Homeland Security Agency and received Enhanced Plan status which allows the state to receive additional Hazard Mitigation Grant Program funds in the aftermath of a disaster.

The plan identifies 13 natural hazards that impact the state, assesses the vulnerability of those hazards, and outlines a strategy to reduce those vulnerabilities. The 13 natural hazards include:

- Hurricane Wind
- Coastal Hazards
- Wind
- Severe Weather
- Tornadoes
- Inland Flooding
- Severe Winter Weather
- Drought
- Wildfire
- Earthquake
- Geologic Hazards
- Dam Failures
- Extreme Heat

All of these hazards and their associated vulnerabilities were carried forward into the RIP and the strategies to reduce vulnerabilities were considered when developing the plan policy and strategy recommendations.

State Freight Plan

The Georgia Freight Plan Update, completed in January 2023 by GDOT and approved by FHWA in April 2023, documents freight planning activities and investments in the state, identifies and

assesses current and future freight opportunities and challenges, and guides freight-related transportation decisions and investments.

The plan discusses the importance of real-time data to enhancing the resilience of freight infrastructure for both civilian and military movements, and the creation of the GDOT Resiliency Committee to help determine which projects to build resilience into and to expand the resilience program.

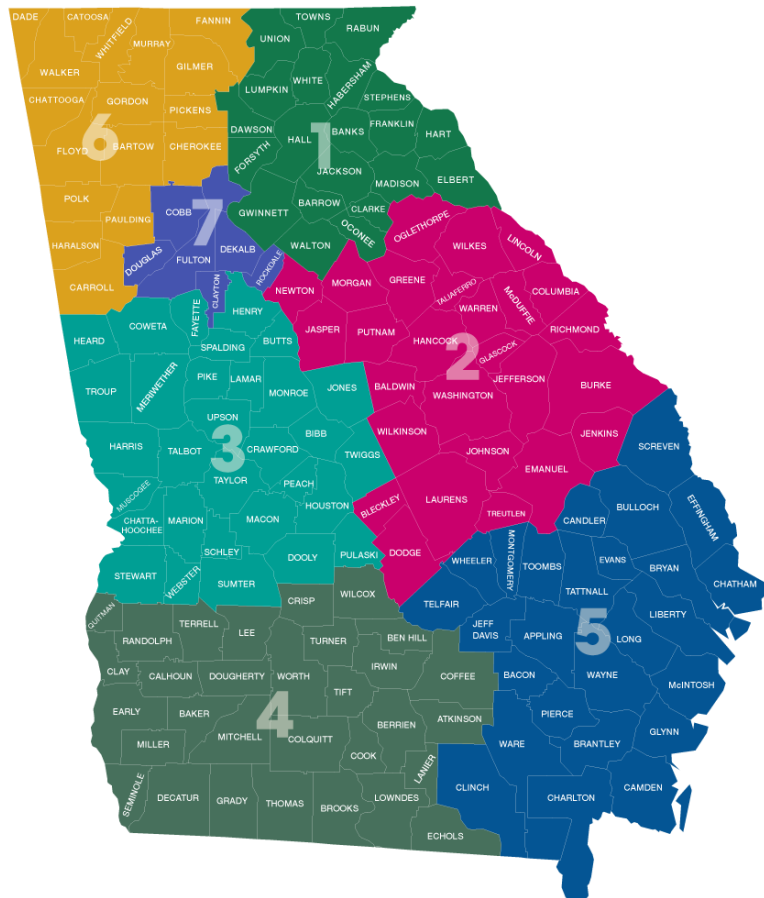
The plan also highlights some of the policies and procedures GDOT has enacted to improve resilience to extreme weather and natural disasters including coordinating with Georgia Tech on a resiliency focused project, creating a Manager of Emergency Operations position, development of guidebooks, designation of hurricane evacuation routes, use of a Road Weather Information System (RWIS), and development of a project using Unmanned Aerial Vehicles (UAV) to assess damage after severe weather events.

The Georgia Tech resilience research project, *Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management*, is part of GDOT's overall research and development efforts focused on resilience, and incorporated evacuation routes and state freight corridors into the criticality factors used to determine a project's criticality score.

Emergency Operations Plan

The Georgia Emergency Operations Plan (EOP), developed and updated by the Georgia Emergency Management and Homeland Security Agency, identifies hazards and critical state facilities. It also dictates emergency management roles and responsibilities for the State Disaster Response Team, including executives, state emergency management personnel, Private-Sector Partners, Non-Governmental Organization partners, local emergency managers, faith-based organizations and any other individuals or organizations expected to support disaster response efforts through emergency management functions. The plan is activated and implemented when an emergency or disaster has occurred or is imminent.

In addition to the statewide EOP, each of the GDOT Districts have their own response plans. The 2050 SWTP/2021 SSTP Resiliency Analysis notes that Districts 2, 3, 4, and 5 each have their own Hurricane Response Plans due to their proximity to the coast.



Source: GDOT

Transportation Asset Management (TAM) Plan

Georgia's TAM Plan defines the condition of the state's pavement and bridge structures (including culverts longer than 20 feet) on the National Highway System (NHS). The plan also outlines the priorities and strategies to manage and preserve these assets over the next 10 years. Section 5 of the plan focuses on risk and resilience and outlines a risk management process adopted by GDOT. The section identifies and rates risks, identifies actions for each risk (including a management plan for top priority risks), and assesses assets repeatedly damaged by emergency events.

2022 Georgia Hazard Identification and Risk Assessment (HIRA)¹⁰

The HIRA, approved and implemented by the Georgia Emergency Management and Homeland Security Agency, provides a statewide overview of how various natural, human-caused, and technological hazards impact the State of Georgia. The assessment identifies and groups hazards and explains how and why each hazard was identified. It details the risk assessment process and the methods used to rank hazard risks; the vulnerability analysis that measures the level of assets, populations, and resources within a given geography; and the consequence

¹⁰ <https://gema.georgia.gov/document/document/2022-hira/download>

analysis that investigates the impact of each hazard on various infrastructure elements. The document ends with an overall assessment and summary of the individual hazard analyses.

Existing Resilience Efforts and Policies

State Efforts

GDOT actively manages the roadway networks and has several existing policies, technologies, and teams which increase its ability to respond and reduce the impact of hazard events. Several GDOT offices undertake information sharing related to traffic, weather, and hazardous conditions to ensure a coordinated approach in providing data and updates critical to the safety of the traveling public. A summary of these efforts and policies is provided below.

Traffic Management Center

GDOT operates the Traffic Management Center (TMC) which actively monitors roadway and intersections via closed-circuit television and aerial drone imagery. This system is in place throughout the year but is also activated to route traffic more effectively during emergency events. This information is also paired with driver information services such as 511, dynamic messaging signs, and the GDOT HERO and CHAMP programs which respond to crashes and disabled vehicles. During major hazard events, the TMC serves as emergency operations back up to the One Georgia Call Center which functions as the State Operations Center (SOC).

Road Weather Information Sensors

GDOT installed 56 Road Weather Information Sensors (RWIS) across the roadway network that are used to record and predict road conditions, appropriate responses (brine teams, routing, etc.) and to convey traveler information.

Statewide Evacuation Route System

GDOT maintains a statewide evacuation route system for the movement of people during emergency events.¹¹ During major hurricane events, GDOT may engage in contra-flow travel along I-16 between Savannah and US 441 in Dublin. This contra-flow travel increases the westbound capacity by utilizing the eastbound travel lanes along I-16.

In the case of a coastal emergency such as a hurricane, the I-16 corridor between Savannah and US 441 in Dublin will only permit westbound traffic. To support this change in traffic flow three median cross overs have been developed; two are positioned east of I-95 to move traffic onto the contra-flow lanes and a third is portioned between Macon and Dublin to divert the contra-flow traffic back to the westbound lanes.

BridgeWatch Software

BridgeWatch® is a web-based monitoring software solution that empowers bridge owners to predict, identify, prepare for, manage, and record potentially destructive environmental events.

¹¹ <https://www.dot.ga.gov/GDOT/Pages/Hurricane.aspx>

The software allows bridge owners to proactively monitor, in real time, bridge infrastructure to better protect against hazardous, costly, and potentially catastrophic events.¹²

GDOT uses BridgeWatch software to monitor scour vulnerable bridges and provides alerts during flooding events. The software is also used during biennial bridge inspections by GDOT bridge inspectors to monitor the condition of bridges and culverts and aid in management of these assets.

Coordinated Highway Assistance & Maintenance Program (CHAMP)

GDOT is the first state to provide complimentary statewide interstate highway assistance for motorists across Georgia. The program, which was created in 2017, has CHAMP operators who provide emergency response, offer motorists assistance, and remove roadway debris and address maintenance issues when reported by motorists by calling 511.

Highway Emergency Response Operators (HERO)

HERO units provide assistance in traffic related incidents in the metro Atlanta area with the primary task to clear roadways so that normal traffic patterns remain intact. HERO units also respond to stranded motorists to provide assistance with difficulties such as flat tires or mechanical issues.

Georgia Emergency Management and Homeland Security (GEMA/HS)

GDOT and GEMA/HS coordinate closely in the preparation for, response to, and recovery from the effects of a disaster on the transportation infrastructure in the state. GDOT provides information to GEMA/HS on impacts and status of the transportation infrastructure, including traffic monitoring during evacuations.

Communications

GDOT also engages with multiple communications providers to not only ensure redundancy within its operations systems but the ability to restore or establish temporary connections during hazard events. These connections include fiberoptic, radio, and cellular networks to provide service throughout the year and during emergencies.

Repetitive Loss Evaluation

As a part of FAST Act requirements, GDOT conducted an evaluation of facilities that have required repairs to emergency events on two or more occasions as part of the development of the Transportation Asset Management Plan¹³. While the evaluation is required on the NHS at a minimum, GDOT undertook the analysis for all state routes.

Biannual Pipe Inspection

GDOT conducts a biannual pipe inspection to identify damaged or non-functioning drainage assets.

¹² <https://usengineeringsolutions.com/bridgewatch/>

¹³ <https://www.tam-portal.com/wp-content/uploads/sites/12/2023/01/GDOT-TAMP-2022.pdf>

Other Efforts

In response to some of the emergency situations encountered over the last several years, GDOT has also implemented some additional policies to enhance their response. These include the following:

- Creation of a GDOT Manager of Emergency Operations position
- Brine application and pre-treating roadways in anticipation of ice
- GDOT disaster response teams have state patrol escorts
- Unification of state agency communications and responses
- Annual real time dry runs for emergency situations for all state agencies involved

Local and MPO Efforts

Resiliency efforts have also been undertaken at the local level by MPOs, counties, and municipalities that contribute to making the transportation system more resilient. Some of these local efforts are documented below.

Camden County 'Rise Ready' Resilience Project

The 'Rise Ready' Resilience Project is focused on raising awareness of flood risks through education and outreach programs as well as the creation of online tools to help users visualize, understand, and plan for flood risks in their communities. The project is a partnership between The Nature Conservancy, Camden County, the Coastal Regional Commission of Georgia, the Georgia Department of Natural Resources, the City of St. Marys, The University of Georgia Marine Extension, and Georgia Sea Grant.

Tools developed as part of the project include a Flood Risk Application, a Community Planner Application, and a Community Rating System (CRS) open space explorer app.

Chatham County Smart Sea Level Sensors Project

The Smart Sea Level Sensors project is a partnership between Chatham Emergency Management Agency officials, City of Savannah officials, and Georgia Tech scientists and engineers who are working together to install a network of internet-enabled sea level sensors across Chatham County. The real-time data on coastal flooding will be used for emergency planning and response.

Atlanta Regional Commission Vulnerability and Resiliency Framework

In 2018, the Atlanta Regional Commission developed a vulnerability and resiliency framework for the Atlanta Region and associated recommendations that can be used as part of a future system vulnerability assessment.

Academic Efforts

Georgia is home to several nationally recognized research institutions, one of which is the Georgia Institute of Technology (Georgia Tech). Georgia Tech leads the GDOT research efforts on resilience and partners with other organizations to improve resiliency in the state.

Georgia Institute of Technology

In addition to work on the local projects mentioned above, Georgia Tech is leading a coastal infrastructure and resilience research initiative called the Georgia Coastal Equity and Resilience

(CEAR) Hub. The new Hub will promote the resilience and sustainability of vulnerable and marginalized Georgia coastal communities through integrative community-driven research, education, training, and outreach to produce equitable, evidence-based, scalable solutions to challenges related to flooding and sea level rise.

Additionally, Georgia Tech partnered with GDOT on a three-year research and development project: *Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management*. The findings and outputs of this research were used to prioritize projects eligible for PROTECT funds. Details on the research and the resilience tool developed as a part of this research are discussed in later sections of the plan.

Existing Laws

Multiple laws exist at the state level related to improving resilience and mitigating the impact of hazardous events. **Table 2** lists those laws and provides a brief description of associated resilience efforts.

Table 2: Existing State Laws Associated with Resilience

Law	Description
Georgia Coastal Management Act (O.C.G.A. §12-5-320)	Authorized the creation of the Georgia Coastal Management Program. Some accomplishments of the program include developing tools for community-based coastal resource management, providing guidance and direct technical assistance for the development of plans that promote resilient development, and supporting pilot projects to make shorelines more resilient.
Georgia Coastal Marshlands Protection Act (O.C.G.A. §12-5-280)	Requires a permit to fill, alter, or locate a structure over marshlands. Established a coastal marsh buffer of 25 ft, prohibiting land-disturbing activities in the buffer area without a variance.
Georgia Mountain and River Corridor Protection Act (O.C.G.A. §12-2-8)	Requires local governments to protect a natural vegetative buffer of 100 ft on each side of the stream banks of every perennial river corridor.
Georgia Shore Protection Act (O.C.G.A. §12-5-230)	Regulates activities and structures in jurisdictional beach and shore areas. The regulated area is a 25 ft zone between private beachfront development and the landward reach of the sand dune or from the high tide line on beaches without dunes.
Georgia Safe Dams Act of 1978 (O.C.G.A. §12-5-370 to 385)	Authorized the creation of the Safe Dams Program which is responsible for developing and maintaining an inventory of dams, classifying dams, and ensuring compliance of all regulated dams.
Georgia Planning Act of 1989 (O.C.G.A. 45-12-200, et seq., and 50-8-1, et seq.)	Requires local governments to develop a comprehensive plan to guide activities over a 20-year planning horizon and called for the DCA and DNR to develop minimum requirements for these plans.
Georgia Erosion and Sedimentation Act of 1975 (O.C.G.A. §12-7-1)	Created a comprehensive plan to control soil erosion and to protect vulnerable environmental resources disturbed by excavation, dredging, and other land-disturbing activities. The law also mandates stream buffer protection.
Georgia Emergency Management Act of 1981 (O.C.G.A. §38-3-1)	Created a state emergency management agency and authorized the creation of local organizations for emergency management. Calls for the coordination of emergency management functions and the rendering of mutual aid for dealing with emergencies and or disasters.
Georgia Soil and Water Conservation Districts Law (O.C.G.A. §2-6-20 and §2-6-27)	Established Soil and Water Conservation Districts to encourage soil and water conservation. The Districts provide a way for citizens to set local resource priorities for state and federal assistance programs.
Georgia Environmental Policy Act (O.C.G.A. §12-16-1)	Requires all State agencies and activities to prepare an Environmental Impact Report as part of a decision-making process for all activities that may have an impact on the environment that considers alternatives to the proposed project or activity.
Metropolitan North Georgia Water Planning District Act (O.C.G.A. §12-5-572)	Created the Metropolitan North Georgia Water Planning District (15 counties and 91 cities) to establish policy, create plans, and promote intergovernmental coordination for all water issues in the district; to facilitate multijurisdictional water related projects; and to enhance access to funding for water related projects among local governments in the district area.
Georgia Building Codes (O.C.G.A. §8)	Establishes the minimum standards for design, construction, quality of materials, use and occupancy, safety and maintenance of various types of buildings to protect life, health, property, and public welfare.
Georgia Records Act (O.C.G.A. §50-18-90)	Outlines the process for how members of the public can request and inspect governmental records.
Georgia Forest Fire Protection Act (O.C.G.A. §12-6-80 to §12-6-93)	Establishes organized and coordinated state-wide protection of forest lands against uncontrolled fire and declares that prescribed burning is an effective method of reducing fuel loads and the potential hazards and impact associated with uncontrolled fires.
Georgia Prescribed Burning Act (O.C.G.A. §12-6-145)	Declares and protects prescribed burning as a valuable and necessary forest management tool.

Existing Tools

A variety of tools and data exist at the national level to help communities plan for and implement measures to improve community resilience (see **Table 3**). These tools and data sources are constantly changing, and new resources are continually being developed as these resources update and become available, the state will consider new data and incorporate if determined to be required.

Table 3: National Resilience Tools and Data

Tool/Data	Agency	Description
National Risk Index	Federal Emergency Management Agency (FEMA)	Online mapping application that identifies communities most at risk of 18 natural hazards. This national application visualizes natural hazard risk metrics and includes data about expected annual losses from natural hazards, social vulnerability, and community resilience. Data is available at the county and census tract level.
Digital Flood Insurance Rate Maps (DFIRMs)	FEMA	A DFIRM is the official digital map of a community on which FEMA has delineated the special flood hazard areas. The map is a computer-based analytical tool that can be used for automated analysis and map updates and is available for many of the nation's highest flood risk areas.
Resilience Analysis & Planning Tool (RAPT)	FEMA	GIS planning tool that helps communities visualize and analyze data to inform strategies for emergency preparedness, response, and recovery. The tool includes pre-loaded data (People & Community, Infrastructure, and Hazard & Risk Indicators) and analysis tools.
Sea, Lake, and Overland Surges from Hurricanes (SLOSH) Model	National Oceanic and Atmospheric Administration's (NOAA) National Weather Service (NWS)	A computerized numerical model used by the National Hurricane Center to estimate storm surge heights, forecast storm surge, and model storm surge vulnerability.
SLOSH Display Program (SDP)	NOAA's NWS	Software provided to emergency managers and other users to visualize the data produced by the National Hurricane Center.
U.S. Climate Resilience Toolkit	NOAA	Website designed to help people find and use tools, information, and subject matter expertise to build climate resilience.
Spatial Hazards Events and Losses Database for the United States (SHELDUS)	Arizona State University	County-level hazard data set for the U.S. that covers natural hazards such as thunderstorms, hurricanes, floods, wildfires, and tornados as well as perils such as flash floods, heavy rainfall, etc. The database contains information on the date of an event, affected location (county and state), and the direct losses caused by the event (property and crop losses, injuries, and fatalities) from 1960 to present.
National Centers for Environmental Information (NCEI)	NOAA	Nation's leading authority for environmental data and manages one of the largest archives of atmospheric, coastal, geophysical, and oceanic research in the world.

Tool/Data	Agency	Description
Digital Coast	NOAA	Website that provides coastal data as well as tools, training, and information to make the data useful. Tools include Climate Mapping for Resilience and Adaptation Assessment Tool, Coastal Resilience Evaluation and Siting Tool, and Coastal Change Hazards Portal.
Climate.data.gov	US General Services Administration (GSA)	Provides data related to climate that can help inform and prepare communities, businesses, and citizens. Also includes data and resources related to coastal flooding, food resilience, water, ecosystem vulnerability, human health, energy infrastructure, and transportation.
Sea Level Tracker	US Army Corps of Engineers (USACE)	Shows actual sea level vs. the project sea level change and depicts the rate of sea level change currently being observed at the selected gauge.
U.S. Drought Portal	NOAA (NIDIS and NCEI)	The U.S. Drought Portal is a website that serves as a one-stop shop for data, decision-support products, resources, and information on drought.
North American Drought Monitor (NADM)	NOAA	A cooperative effort between drought experts in Canada, Mexico, and the US to monitor drought across the continent. Updated monthly.
Vulnerability Assessment Scoring Tool (VAST)	US Department of Transportation (US DOT)/FHWA	An excel based tool used to identify the vulnerability of transportation projects based on project type and identified environmental stressors. Vulnerability is determined by the exposure, sensitivity, or adaptive capacity of the transportation asset.
National Bridge Inventory (NBI)	FHWA	Data on the location, description, classification, and general condition of the nation's bridges located on public roads, including interstate highways, state and county roads, and publicly accessible bridges on federal and tribal lands.
Overall Condition Index (OCI)	USACE	Benchmark used to measure the general condition of publicly maintained streets.
The Climate Explorer	The National Environmental Modeling and Analysis Center (NEMAC) at UNC-Asheville	Interactive tool that provides graphs, maps, and downloadable data of observed and projected climate conditions for every county in all 50 states and U.S. territories.
Social Vulnerability Index (SVI)	Center for Disease Control (CDC) Agency for Toxic Substances and Disease Registry (ATSDR)	A tool that identifies the social vulnerability uses 16 U.S. census variables to help local officials identify communities that may need support before, during, or after disasters
Real-time Data	United States Geological Survey (USGS)	Continuously gathers and measures current conditions related to earthquakes, volcanoes, floods, wildland fires, landslides, and drought.
Pipeline Incident Database	Pipeline and Hazardous Materials Safety Administration (PHMSA)	An extensive, public pipeline database available to states and industry to identify and/or assess pipeline and underground natural gas storage safety challenges and potential solutions.

Tool/Data	Agency	Description
DamWatch®	USDA Natural Resources Conservation Service (NRCS)	Web-based application developed to assist with monitoring and managing NRCS-assisted dams by provided real-time updates of rainfall, snowmelt, stream flow and seismic events that could affect dam safety.

Climate Vulnerability, Risk, and Resilience Assessment

The research sponsored by GDOT and conducted by Georgia Tech reviewed climate vulnerability, risk, and resilience of Georgia's pavements, bridges, and culverts using the Spatial Hazard Events and Losses Database for the United States (SHELDUS) data¹⁴. This dataset was chosen because it is updated annually, and it was the primary data source used in the GDOT 2050 SWTP/2021 SSTP Resilience Analysis. In addition to supporting ongoing resilience efforts and this Resilience Improvement Plan, the results of the assessment will also be incorporated in the Georgia Statewide Travel Demand Model. SHELDUS data was used as a proxy for future conditions for the development of the RIP with an assumption that extreme weather events are becoming more frequent and severe. GDOT is in active conversation with their academic partner (Georgia Tech) and consultant team to incorporate considerations for future climate conditions in the assessment of vulnerability, climate risks, and resilience in Georgia's transportation system.

Climate Projections and Proactive Adaptation

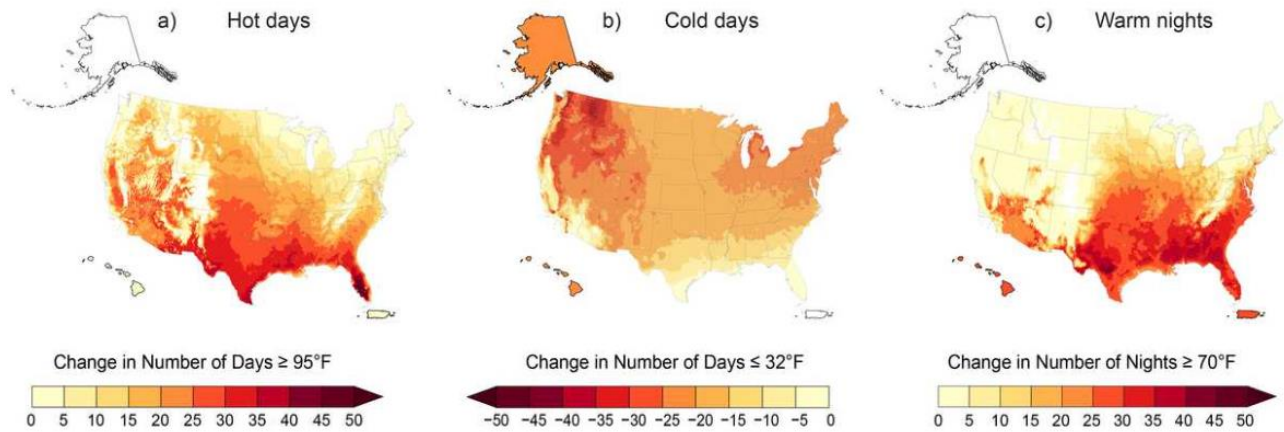
GDOT is currently performing research to understand projected precipitation and temperature changes to estimate the future impacts of extreme weather to transportation assets. Based on the Fifth National Climate Assessment (NCA5), published in 2023, the US is projected to warm faster than the global average, and precipitation changes are projected to be larger at higher levels of warming ([Fifth National Climate Assessment \(globalchange.gov\)](https://www.globalchange.gov/)).

Figure 5 and **Figure 6** illustrate the projected change in temperature and precipitation assuming a 2°C (3.6°F) increase in global temperatures as compared to the time period between 1991 and 2020. The analysis indicated there will be more hot days and warm nights (**Figure 5**), particularly in the southeastern states (Georgia included). Similarly, increases in the frequency and severity of heavy precipitation in the southeast is expected assuming the same global temperature increase (**Figure 6**).

Further analysis as part of the climate assessment indicates that sea levels have risen along most US coastlines between 1993 and 2020. The highest rates of sea level rise have occurred along the Gulf Coast and the Atlantic Coast as illustrated **Figure 7**. Continued sea level rise and coastal flooding will harm rapidly growing communities and transportation assets in the Southeast.

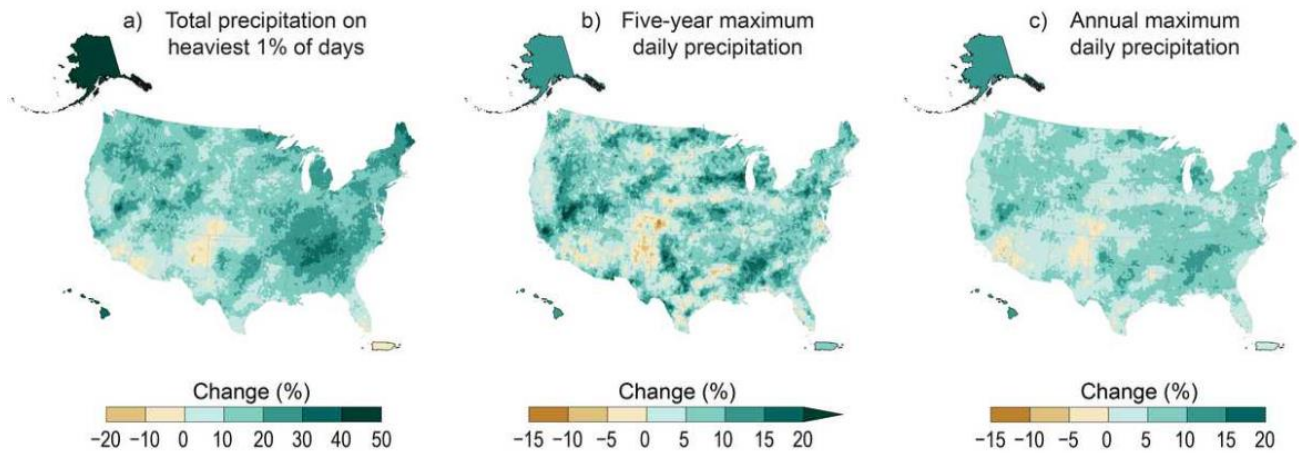
¹⁴ The research involves a risk-based assessment of different asset types on the transportation system (bridges, pavement, and culverts). The word "risk" implies a probability of something happening in the future based on the current and/past patterns (e.g. SHELDUS data). The following section provides details on the factors that went in the calculation of risk, and this accounts for future weather events' impact on the transportation system.

Figure 5 Projected Changes to Temperature Extremes at a 2°C (3.6°F) Increase in Global Temperatures



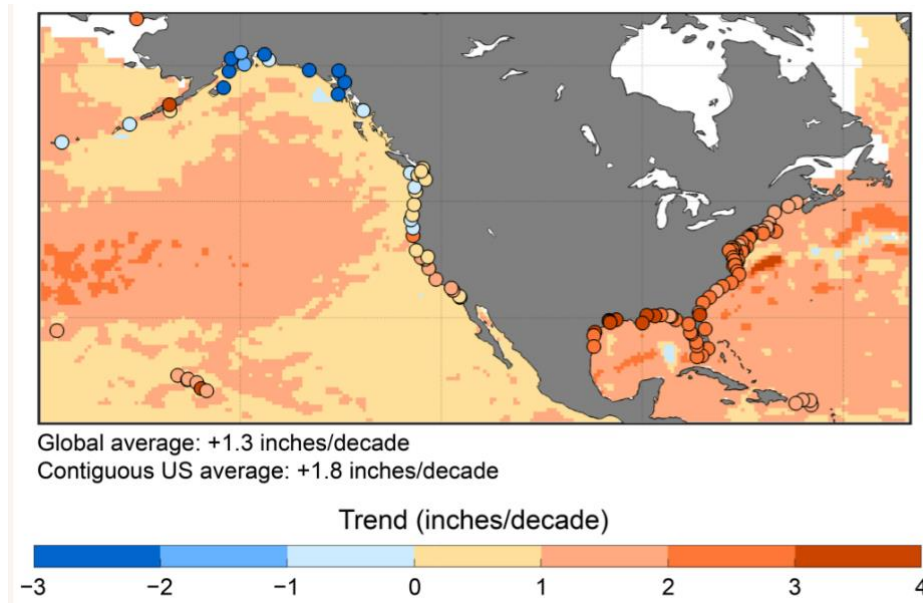
Source: Climate Trends (globalchange.gov)

Figure 6 Projected Changes to Precipitation Extremes at a 2°C (3.6°F) Increase in Global Temperatures



Source: Climate Trends (globalchange.gov)

Figure 7 Sea Level Trends (1993-2020)

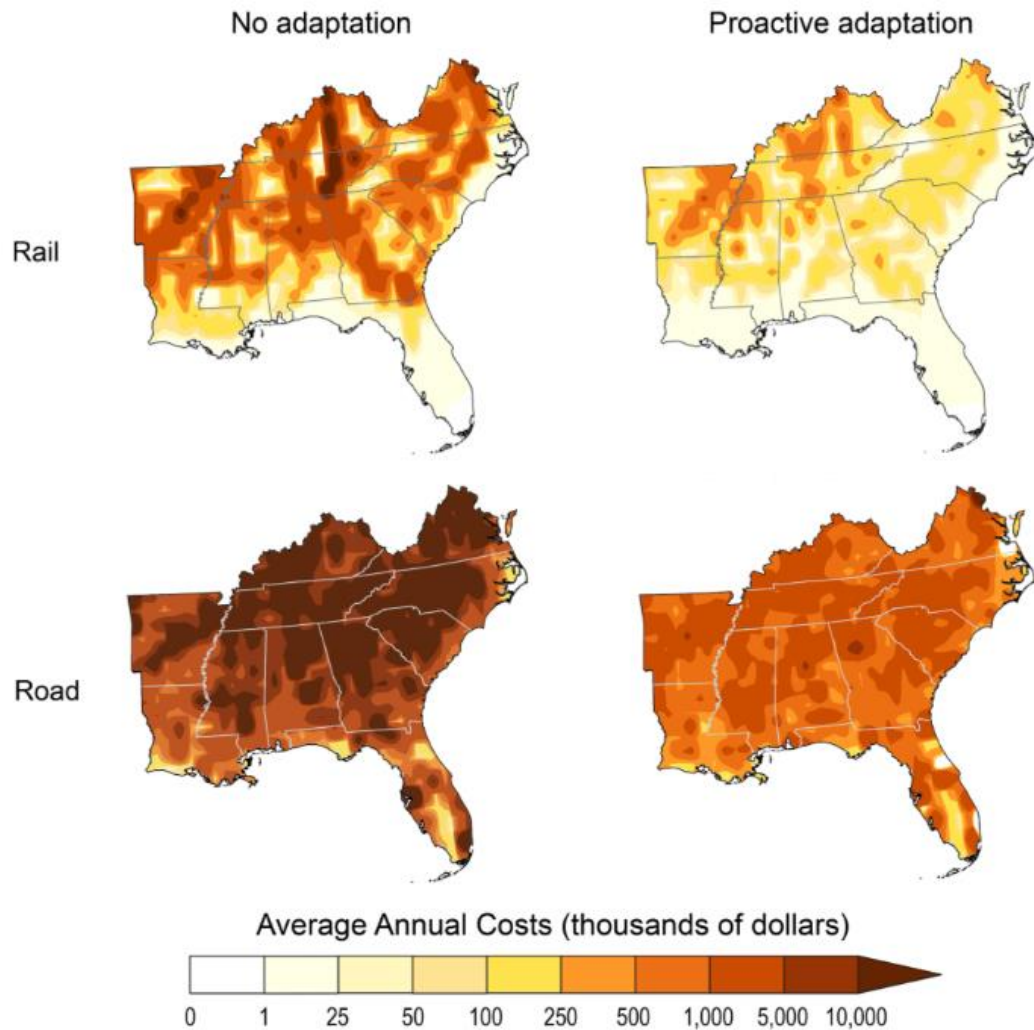


Source: Climate Trends (globalchange.gov)

Meanwhile, proactive adaptation to temperature and precipitation changes and sea level rise could offset millions of dollars in future transportation infrastructure costs. **Figure 8** shows projected transportation infrastructure damages and the benefits of proactive adaptation in 2050 under an intermediate climate change scenario. For example, the left figures show the additional annual average system costs (as compared to 1986–2005 in 2018 dollars) to rail infrastructure (**top**) and road networks (**bottom**) in 2050, assuming no adaptation. Proactive adaptation such as predicting climate risks and making upfront investments to strengthen these systems before damage occurs could significantly reduce these costs (**right figures**). Proactive adaptation strategies could include temperature sensors for railroad tracks and working to reduce disruption times for roads undergoing repairs ¹⁵.

¹⁵ <https://nca2023.globalchange.gov/chapter/22/#fig-22-15>

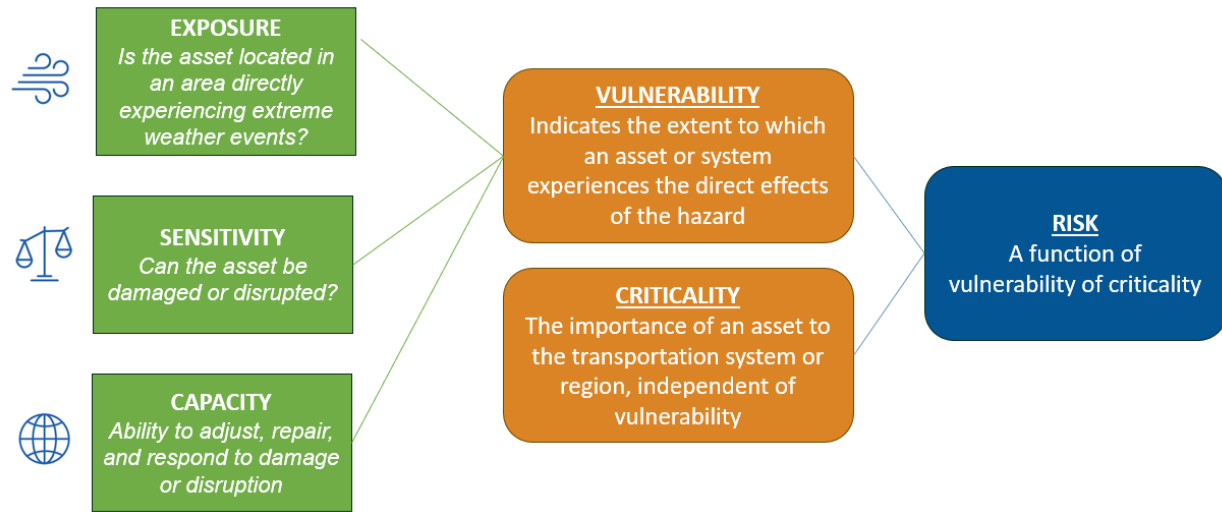
Figure 8 Projected Future Transportation Infrastructure Costs by Proactive Adaptation



Source: Climate Trends (globalchange.gov)

To proactively adapt to such temperature/precipitation change and sea level rise, GDOT worked with a research team from Georgia Tech to develop a tool that can support decision making based on the criticality, vulnerability, and climate risks of the state's transportation infrastructure and assets. This tool is called the multi-hazards exposure, vulnerability, and risk analysis (MHEVRA). Vulnerability is a function of asset exposure, asset sensitivity, and adaptive capacity in the tool. Vulnerability and criticality in turn determine the level of climate risk. **Figure 9** shows relationships among these key resilience components.

Figure 9: Relationship Between Key Resilience Components



The following section includes descriptions of the four resiliency factors used to determine dynamic risk:

- Asset Exposure
- Asset Sensitivity
- Adaptive Capacity
- Asset Criticality

Asset Exposure

Asset exposure refers to whether an asset or system is located in an area experiencing direct effects of climate variability and extreme weather events. A multi-hazards exposure index was developed based on the cumulative number of major climate/weather-related events that have occurred in each county and GDOT district¹⁶ and the level of hazard activity by hazard type.

The exposure index is combined from hazard scores resulting from a GIS-based hot spot analysis and weights capturing the specific hazards that have the potential to reduce asset performance, that is, the hazard-asset pairs that have known failure mechanisms.

Figure 10 shows the results from the hot spot analysis and **Figure 11** shows the hazard-asset pairs that have failure mechanisms. Equation 1, found on Page 28, shows how the multi-hazards exposure score for pavement is calculated (a similar approach is used for bridges and culverts).

¹⁶ The cumulative count of major climate/weather-related events was based on the number of \$billion hazard events that occurred between 1960-2020.

Figure 10: Climate Hazard Exposure - Score Categories based on Hot Spot Analysis Results

Hot Spot	Definition	Score	Normalized Score
Intensifying (I)	A location that has been a statistically significant hot spot for more than 90% time steps, including the final time step. In addition, the intensity of clustering of high counts in each time step is increasing.	8	1.00
Persistent (P)	A location that has been a statistically significant hot spot for more than 90%, with no discernible trend indicating an increase or decrease in the intensity of clustering over time.	7	0.875
Diminishing (D)	A location that has been a statistically significant hot spot for more than 90% time steps. In addition, the intensity of clustering of high counts in each time step is decreasing, or the most recent time step is not hot.	6	0.75
New (N)	A location that is a statistically significant hot spot only for the last time steps of the time series.	5	0.675
Consecutive (C)	A location with a single uninterrupted run of statistically significant hot spot bins in the final time-step intervals. The location has never been a statistically significant hot spot prior to the final hot spot run and less than 90% of time steps are statistically significant hot spots.	4	0.5
Sporadic (S)	A location that is an on-again then off-again hot spot. Less than 90% time steps have been statistically significant hot spots.	3	0.375
Oscillating (Os)	A statistically significant hot spot for the final time-step that has a history of also being a statistically significant cold spot during a prior time step. Less than 90% of the time-step intervals have been statistically significant hot spots.	2	0.25
Historical (H)	The most recent time period is not hot, but at least 90 percent of the time-step intervals have been statistically significant hot spots.	1	0.125
Others (Ot)	All cold spots and other non-significant hot-spot categories.	0	0

Source: Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management Draft Final Report

Figure 11: Climate Hazard Exposure - Weights from Hazard-Asset Pairs with Failure Mechanisms

HAZARD	ASSET	Pavement	Bridge	Culvert
Tornado				
Inland Flooding				
Hurricane				
Severe Storm				
Hail				
Lightning				
Coastal Flooding				
Drought				
Severe Winter Weather				
Wildfire				
Wind				
Extreme Heat				
Fog				
Landslides				

Legend

Asset has notable response/reduced performance to hazard

Asset does not have notable response/reduced performance to hazard

Source: Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management Draft Final Report

$$\text{General Exposure Score for Pavements } (E_p) = \sum_{i=1}^n W_i \times e_{ic} \dots \quad \text{Equation 1}$$

where E_p = General Exposure Score for Pavements (capturing pavement exposure to all climate hazards that affect pavement performance, and factoring in the relative importance of those hazards (individually and combinatively) to pavement performance)

W_i = Weight of particular Hazard i

S_i = Normalized Hot Spot Score for a particular Hazard i (based on the hot spot category identified in Table 2; the hot spot category is identified based on the hot spot analysis for the county where the infrastructure asset is found.)

By default, equal weights are assigned to all the hazards that can result in reduced performance of a particular asset (that is, pavement, bridge, or culvert). The weights may be modified based on the knowledge of local practitioners of the relative importance of specific hazards, as well as their interactions, to asset and system performance.

Asset Sensitivity

Asset sensitivity refers to whether the asset may be damaged or disrupted by a stressor. The more sensitive an asset, the more likely it is to be damaged or disrupted by a stressor.

Asset sensitivity was determined by the condition of the asset. The condition of pavement was determined using the Pavement Condition Index (obtained from GDOT), bridges using the Bridge Sufficiency Rating, and culverts using the Culvert Condition Index (bridge and culvert data was obtained from the National Bridge Inventory)¹⁷.

Adaptive Capacity

Adaptive capacity is the ability of a system to adjust, repair, and respond to damage or disruption. Adaptive capacity may be measured in various ways. In their assessment, the Georgia Tech Research Team proposed using a Resilience Rating Tool and Scorecard patterned after the World Bank's Resilience Rating System.¹⁸

It was assumed that all transportation districts were at the beginning stages of building adaptive capacity. As a result, the Research Team assigned the same unit score to all seven GDOT districts. The Georgia Tech Research Team suggests more detailed self-assessments in the future (at the district or local level) that consider various measures of adaptive capacity including route redundancy.

This component is expected to, and should continue to evolve, during future resilience assessments.

Asset Criticality

Asset criticality refers to the importance or value of infrastructure asset, in terms of the cost to users, owners, and society from a loss in function.

¹⁷ <https://www.fhwa.dot.gov/bridge/nbi.cfm>

¹⁸ <https://documents1.worldbank.org/curated/en/701011613082635276/pdf/Summary.pdf>

The asset criticality score is combined from a score using GDOT State Route Prioritization Criteria shown in **Figure 12** (and weighted at 80 percent) and the social vulnerability index¹⁹ (weighted at 20 percent) to incorporate social equity considerations into the assessment.

Figure 12: Factors for Estimating Criticality Score

Criteria	Criticality Score			
	1 Low Impact	2 Moderate Impact	3 High Impact	4 Very High Impact
AADT	Low	Medium	High	
Functional Classification of Roads	Unclassified Routes	U.S. Highways	U.S. Highways, Intermodal Corridors	U.S. Highways, Intermodal Corridors, Interstates
Number of Lanes	<4	>=4		
Governor's Road Improvement Program			Yes	
Evacuation Route			Yes	
STRAHNET/ STRAHNET Connectors				Yes
State Freight Corridors				Yes

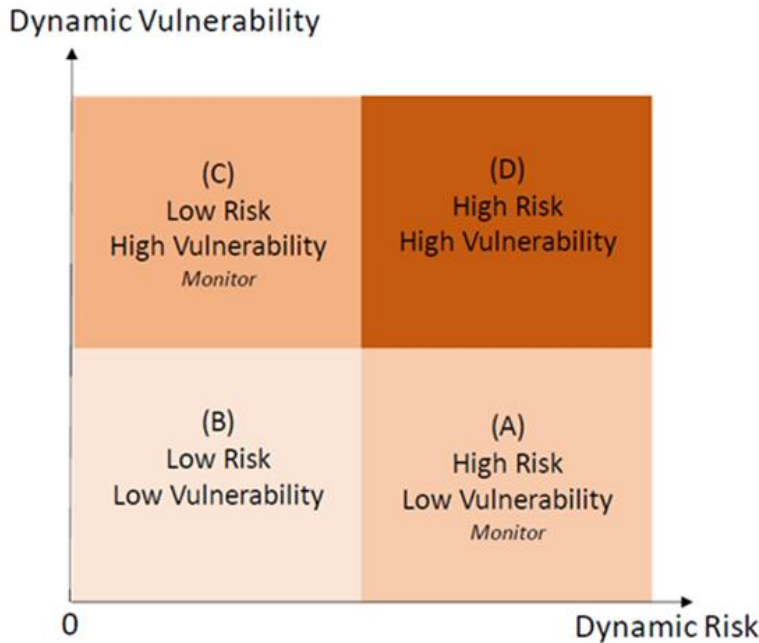
Source: Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management Draft Final Report

Asset Prioritization

The Georgia Tech Research Team used a matrix to prioritize the assets and other system elements (e.g., corridors) based on the relative urgency of need for more detailed examination to address their general vulnerability to climate events (see **Figure 13**). The color gradient signifies dynamic risk, which is the product of criticality and dynamic vulnerability.

¹⁹ The social vulnerability index (SVI) is a tool developed by the Centers for Disease Control and Prevention (CDC) and the Agency for Toxic Substances and Disease Registry (ATSDR). The tool refers to the potential negative effects on communities caused by external stresses on human health – including natural or human-caused disasters or disease outbreaks.

Figure 13: Asset Prioritization Matrix



Source: Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management Draft Final Report

Assessment Results

The RIP relies on a risk-based assessment of the three asset types: (1) bridges, (2) pavement, and (3) culverts, which are the main components of the surface transportation system. It aims to build resilience by prioritizing investments where the risk is higher to the asset. By doing so, it benefits any modes that are running on the surface transportation system, highway or non-highway, including transit and rail. GDOT will continue to explore additional opportunities for engaging a broader group of stakeholders (including transit representatives such as MARTA and the ATL Authority) in resilience discussions and assessing vulnerabilities of additional asset types beyond bridges, pavements, and culverts through the Resiliency Committee.

The following section summarizes results for vulnerability and risk assessment.

Vulnerability

Vulnerability is defined as the degree to which a system is susceptible to, or unable to cope with adverse effects of extreme weather events. In the transportation context, climate vulnerability is a function of the transportation system's exposure to climate effects, sensitivity to climate effects, and adaptive capacity.

Based on the analysis, the highest percentage of pavement and culverts identified as high vulnerability are located in northern Georgia in Districts 7, 6, and 1 respectively. The highest percentage of bridges identified as high vulnerability are also located in in northern Georgia in Districts 6, 7, and 1, respectively (see **Figure 14**).

Risk

Risk is defined as a combination of the likelihood that an asset will experience an extreme event's impact, and the severity or consequence of that impact. Risk can also be defined as the potential for harm expressed as the product of the probability and the consequence of an event and often expressed in relative term as high, moderate, and low.²⁰

The highest percentages of high-risk pavement are in north Georgia in Districts 7, 6, and 1, respectively. The highest percentages of high-risk bridges and culverts are also located in north Georgia in Districts 7, 6, and 1, respectively (see **Figure 15**).

¹GDOT 2050 SWTP/2020 SSTP Resilience Analysis

Figure 14: Vulnerability Results

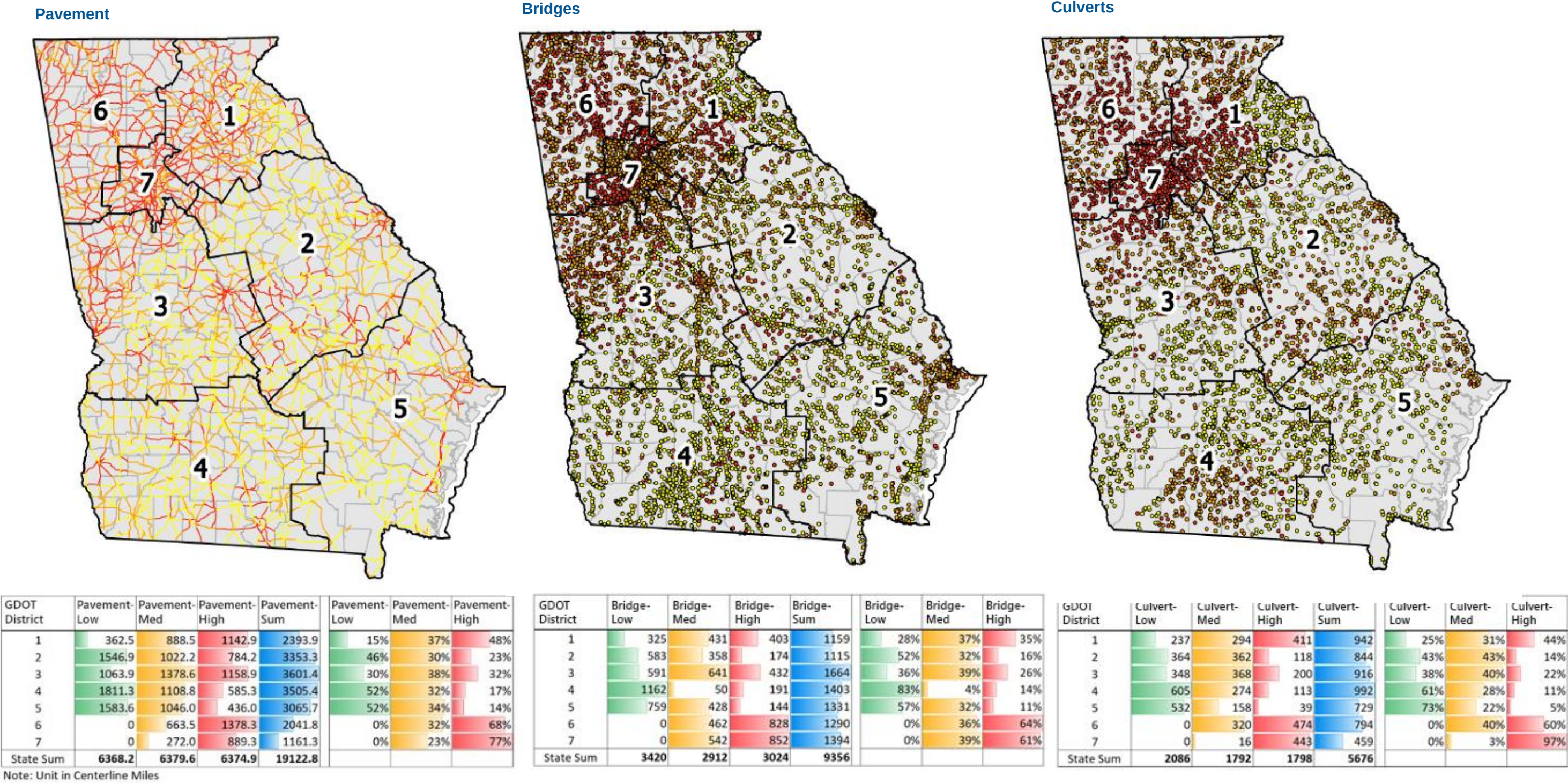
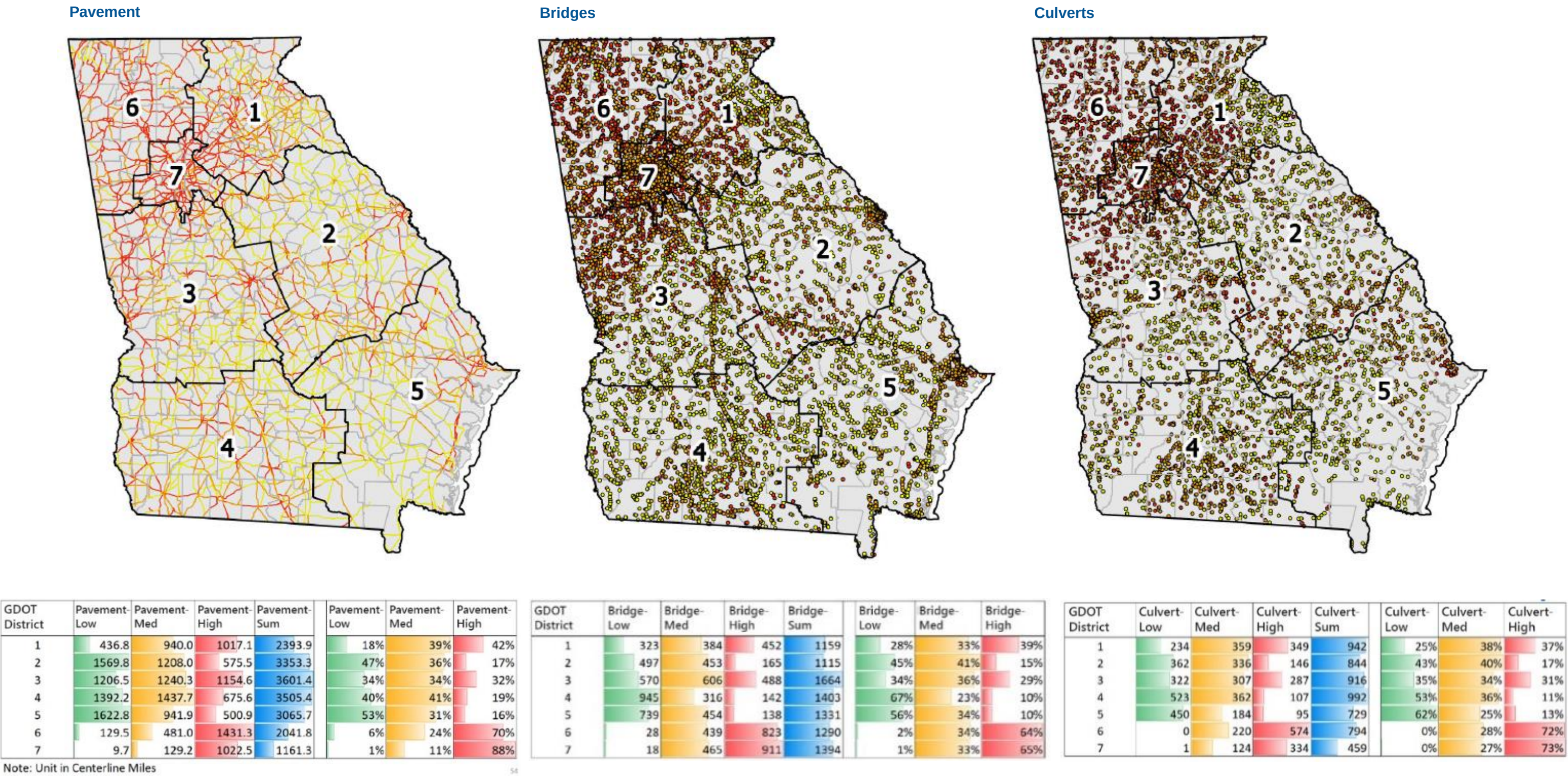


Figure 15: Risk Results



Policy and Strategy Recommendations

As noted in the Existing Resilience Efforts & Policies section, GDOT is already implementing many strategies and actions related to resilience as part of its day-to-day operations. This section highlights potential new or additional strategies GDOT could consider. These strategies were developed in response to the vulnerabilities and potential impacts identified in this plan.

The strategies address all phases of planning and managing the transportation system from planning and project development through design and construction, operations and management, emergency management, and into asset management and maintenance.

Planning

- Monitor and track how resilience is changing over time
 - Monitor and regularly report resilience performance metrics found in **Table 4** on the following page
 - Continually identify potential impacts to the transportation system
 - Monitor assets or portions of the transportation system that fall within the categories of high vulnerability/low criticality or low vulnerability/high criticality
 - Continue to monitor trends in hazardous events to prepare strategies for addressing potential future impacts
- Fill gaps in data availability or analysis capacity to better address potential impacts to the transportation system
 - Analyze assets other than roadways, bridges, and culverts
 - Strengthen adaptative capacity data
 - Incorporate sea level data into the risk assessment
 - Incorporate weather projections into the risk assessment
- Refer to recommendations in the *Incorporating Resilience Considerations in Transportation Planning, TSMO, and Asset Management Draft Final Report* when updating the LRTP, SWTP/SSTP, TAMP, and TSMO Plan
- Conduct and regularly update the vulnerability assessment included in this plan based on updated available evidence based or scientifically proven data and consistent standards
- Incorporate resilience strategies into planning studies and needs assessments

Project Development

- Integrate resilience strategies into project alternatives to support purpose
- Conduct technical analyses of potential hazards as input to the project development process
- Use and build upon the tools developed by Georgia Tech (Adaptive Resilience Capability Maturity Model; Flexibility and Agility Scorecards; and a Resilience Rating Tool and Scorecard) to self-assess and continue to develop adaptive capacity
- Consider the possible use of carbon reduction strategies with resilience strategies to achieve more sustainable resilience in the longer term

Design, Materials, and Construction

- Evaluate project designs considering ability to withstand potential hazards and risks.
- Evaluate higher-quality and more resilient materials and techniques
- Look for opportunities to implement natural infrastructure solutions in project designs
- Evaluate investment strategies that reduce overall operating and response costs to hazards

Traffic Operations and Emergency Management

- Consider hazards and resilience strategies in ITS and TSM&O activities
- Expand use of technology during major events to collect and share information with emergency responders and the public
- Evaluate and enhance processes, procedures, and projects after hazard events
- Coordinate to ensure consistency between this plan and state, regional, and local emergency management plans

Asset Management and Maintenance

- Evaluate infrastructure retrofit and adaptation strategies in Resurfacing, Restoration, and Rehabilitation (3R) projects
- Develop methodologies and tools for analyzing the cost benefits of incorporating resilience in asset management activities
- Utilize more technological capabilities and data in inspections and maintenance of transportation infrastructure and assets

Performance Metrics

Performance metrics were developed based on the plan goals and baseline metrics established from the asset climate and vulnerability and risk assessments completed by Georgia Tech in the GDOT resilience research project. High vulnerability and critical assets were identified and rounded to the nearest whole percent. For example, the percentage of high vulnerability bridges is 32% and the target is no more than 30% of the bridges should be in the high vulnerability category. These performance targets are shown in **Table 4**.

Table 4: RIP Performance Metrics

Performance Measure	Target
Percentage of high vulnerability pavement	30%
Percentage of high vulnerability bridges	30%
Percentage of high vulnerability culverts	30%

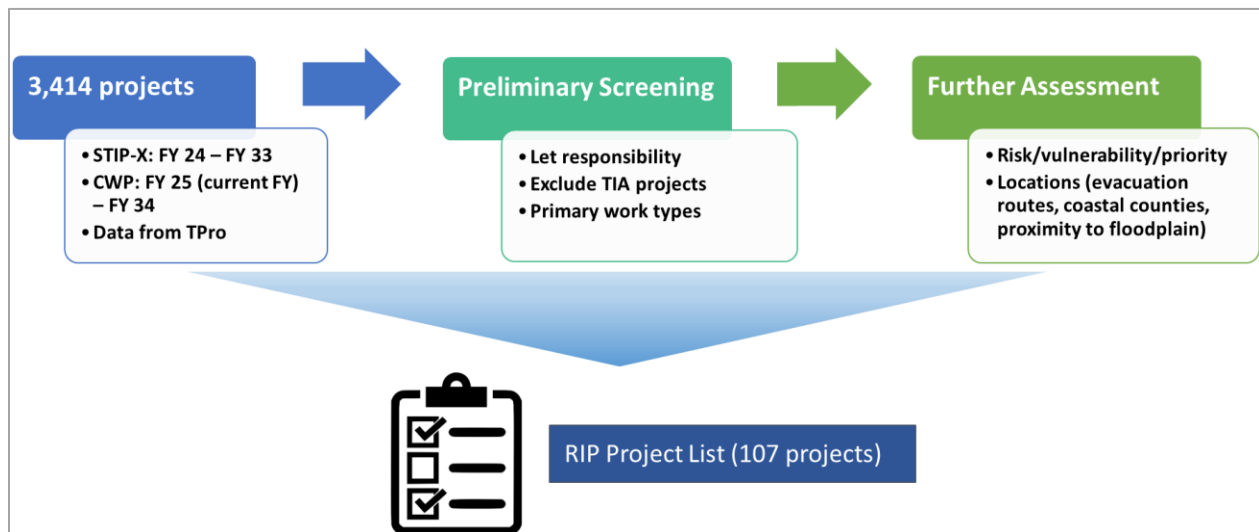
These performance measures are aligned with the foundational, catalytic, and innovation goals identified for ensuring a resilient transportation system and maintaining the nation's leading infrastructure system. With the identification of the high hazard areas and infrastructure with ranked with high vulnerability and high criticality, the risk to the mobility of the state's residents is reduced; and ensures investments on those areas with the greatest vulnerability and highest criticality are met.

Project Screening & Prioritization

Project Screening

All projects between FY 2024 and FY 2034 were screened to be included in the RIP to cover all projects in the current STIP-X, the 10-year period that covers the STIP (FY 24 – FY 27), TIPs, and GDOT's current Construction Work Program (FY 25 – FY 34). As a result, the list of projects included in the RIP should be considered a snapshot in time and will be periodically updated to reflect the current need and the goals for resilience. The general screening approach is shown in **Figure 16** below.

Figure 16: General Screening Approach



Let Responsibility and Regional Priority

Projects that have yet moved to construction and are let by GDOT are preferred. Additionally, Transportation Improvement Act (TIA) projects, projects that are approved for TIA funding by the participating regional commissions, are omitted. These projects are considered regional priorities that already have dedicated funding.

Primary Work Type

The RIP gives a higher priority to projects whose primary purpose is to maintain and rehabilitate transportation assets. These projects are listed in “Tier 1.” Projects in “Tier 2,” which involve capacity and operational improvements, need to be on an evacuation route to be considered eligible. **Figure 17** shows Tier 1 and Tier 2 work type categories used in the screening process.

Due to a large number of bridge projects, only those that are unlet and that were identified by the Office of Bridge Design and Maintenance for the PROTECT funding were included.

Figure 17: Primary Work Types Screening Tiers

Tier 1	Tier 2
Bridges	Passing Lanes
Shoulder Work	Widening
Slope Repair	Interchange
Resurface & Maintenance	Turn Lanes
Grade, Drain, Base & Surface	Signals
Pavement Rehabilitation	Intersection Improvement
Drainage Improvements	ITS
Curb & Gutter	Widening and Resurfacing
Culvert	
Concrete Rehabilitation	

Further Assessment

This step further evaluates projects based on their locations such as coastal counties and FEMA floodplain information to assist in prioritizing investment and/or in applying for PROTECT funding using the FHWA PROTECT Eligibility Screening Form.

Assessment of Criticality, Vulnerability, and Risk based on Project Locations

- Use the Multi-Hazards Exposure, Vulnerability and Risk Assessment (MHEVRA) tool where data is available.
- Prioritize projects using the tool (priority based on criticality/vulnerability/risk) or project types where data is not available from the tool.

Including FEMA Flood Zone information:

- 1% Annual Chance of Flood Hazard (100-year Floodplain)
- 0.2% Annual Chance of Flood Hazard (500-year Floodplain)

Table 5 shows the list of projects recommended for PROTECT funding under the Resilience Improvement Plan as of October 2024.

This screening process excludes projects that were already approved for PROTECT funding by FHWA in FY 23 and FY 24 (**Table 6**).

Table 5: Projects Recommended for PROTECT Funding (FY 24 – FY 34)

No.	Project ID	County	GDOT District	Description	Primary Work Type	Criticality Class	Vulnerability Class	Risk Class	Priority	On Evacuation Route	Coastal County	1% Annual Chance Flood Hazard Area	0.2% Annual Chance Flood Hazard Area
1	0008376	Muscogee	3	I-185 FROM CR 62/OLD CUSSETA ROAD TO SR 22 SPUR	Shoulder Work	3	3	3	First	No	No	Yes	Yes
2	0010295	Lowndes	4	I-75 @ SR 376 - PHASE II	Interchange	3	2	3	First	Yes	No	No	No
3	0013390	Lumpkin	1	SR 52 @ CHESTATEE RIVER - BRIDGE REPLACEMENT	Bridges	2	3	3	First	No	No	Yes	Yes
4	0013714	Brooks	4	SR 76/SR 333 @ CS 834/CRAWFORD ST & CSX #636942L IN QUITMAN	Bridges	2	3	3	First	Yes	No	No	No
5	0013745	Glynn	5	SR 25 SPUR EAST @ MACKAY RIVER 3 MI E OF BRUNSWICK	Bridges	3	3	3	First	No	Yes	Yes	Yes
6	0013827	Pickens	6	SR 136 @ TALKING ROCK CREEK 3 MI N OF JASPER	Bridges	2	3	3	First	No	No	Yes	No
7	0014894	Bulloch	5	SR 26 FROM SR 67/SR 73 TO CS 802/LOVETT STREET IN STATESBORO	Widening	2	2	3	First	Yes	No	Yes	Yes
8	0015413	Chattooga	6	SR 1/US 27 @ TAYLORS RIDGE - PHASE II	Slope Repair	3	3	3	First	No	No	No	No
9	0015560	Newton	2	SR 81 @ YELLOW RIVER IN PORTERDALE	Bridges	2	3	3	First	No	No	Yes	Yes
10	0015605	Bulloch	5	CR 927/OLD HWY 46 @ ASH BRANCH 11.6 MI SE OF BROOKLET	Bridges	2	3	3	First	No	No	Yes	No
11	0015632	Coffee	4	CR 705/BRIDGETOWN ROAD @ SATILLA RIVER 11 MI W OF DOUGLAS	Bridges	2	3	3	First	No	No	Yes	No
12	0016514	Echols	4	SR 89/US 441 @ LITTLE SUWANNEE CREEK 8 MI S OF FARGO	Bridges	3	3	3	First	Yes	No	Yes	No
13	0016515	Fannin	6	SR 5 @ TOCCOA RIVER IN MCCAYSVILLE	Bridges	3	3	3	First	No	No	Yes	No
14	0016521	Harris	3	SR 103 @ HOUSE CREEK 6.5 MI SE OF WEST POINT	Bridges	3	3	3	First	No	No	Yes	No
15	0016522	Harris	3	SR 103 @ SAND CREEK 5.5 MI SE OF WEST POINT	Bridges	3	3	3	First	No	No	Yes	No
16	0016527	Meriwether	3	SR 362 @ RED OAK CREEK 6 MI NE OF GREENVILLE	Bridges	2	3	3	First	No	No	Yes	No
17	0016529	Murray	6	SR 61 @ PERRY CREEK 6.3 MI N OF CRANDALL	Bridges	2	3	3	First	No	No	No	No
18	0017222	Carroll	6	SR 113 @ ALLEN CREEK & @ BETHEL CREEK	Bridges	2	3	3	First	No	No	Yes	No
19	0017223	Chattooga	6	SR 114 @ CHATTOOGA RIVER TRIBUTARY	Bridges	2	3	3	First	No	No	Yes	No
20	0017224	Chattooga	6	SR 114 @ MOSTELLER CREEK	Bridges	2	3	3	First	No	No	Yes	No
21	0017415	Chatham	5	SR 26/US 80 @ LAZARETTO CREEK	Bridges	3	3	3	First	No	No	Yes	No
22	0017515	Chatham	5	I-16 @ SR 17 CONN	Intersection Improvement	3	3	3	First	Yes	No	No	No
23	0017729	Dawson	1	SR 53 @ THOMPSON CREEK	Bridges	3	3	3	First	No	No	No	No
24	0017734	Habersham, White	1	SR 384 @ CHATTAHOOCHEE RIVER	Bridges	2	3	3	First	No	No	Yes	No
25	0017756	Bibb	3	I-16 @ I-75 - ITS DEPLOYMENT - PHASE I	ITS	3	3	3	First	Yes	No	Yes	Yes

No.	Project ID	County	GDOT District	Description	Primary Work Type	Criticality Class	Vulnerability Class	Risk Class	Priority	On Evacuation Route	Coastal County	1% Annual Chance Flood Hazard Area	0.2% Annual Chance Flood Hazard Area
26	0017915	Bibb	3	I-16/I-75 INTERCHANGE ITS DEPLOYMENT - PHASE II	ITS	3	3	3	First	Yes	No	Yes	Yes
27	0017929	Burke	2	SR 56 @ ROCKY CREEK	Bridges	3	3	3	First	No	No	Yes	No
28	0017933	Emanuel	2	SR 26 @ FIFTEEN MILE CREEK	Bridges	3	3	3	First	Yes	No	Yes	No
29	0017934	Gilmer	6	SR 52 @ CARTECAY RIVER	Bridges	3	3	3	First	No	No	Yes	Yes
30	0017935	Gilmer	6	SR 52 @ LICKLOG CREEK 6.6 MI SE OF ELLIJAY	Bridges	2	3	3	First	No	No	Yes	No
31	0017969	Whitfield	6	SR 3/US 41 FROM SR 71 TO MILL CREEK	Resurface & Maintenance	3	3	3	First	No	No	Yes	No
32	0018297	Gwinnett	1	SR 120 FROM CR 5529/BOLES FARM LANE TO CR 709/PINE STREET	Drainage Improvements	2	3	3	First	No	No	Yes	No
33	0019093	Grady, Thomas	4	SR 38/US 84 EB @ BARNETTS CREEK & OVERFLOW	Bridges	2	3	3	First	No	No	Yes	No
34	0019286	Chattooga	6	SR 1/US 27 FROM CS 588/WASHINGTON STREET TO SR 48	Drainage Improvements	2	3	3	First	No	No	No	No
35	0019358	Murray	6	SR 61 FROM SR 52 TO CS 603/OLIVE STREET	Drainage Improvements	2	3	3	First	No	No	No	No
36	0019360	Dade	6	SR 136 @ MAGBY GAP	Slope Repair	2	3	3	First	No	No	No	No
37	0019630	Walker	6	SR 1BU FROM SR 193 TO SR 1 IN LAFAYETTE	Pavement Rehab	2	3	3	First	No	No	No	No
38	0019684	Carroll	6	SR 113 @ CR 261/PLEASANT RIDGE ROAD	Resurface & Maintenance	1	3	3	Second	No	No	No	No
39	0020118	Cobb	7	I-75 @ HOPE CREEK	Drainage Improvements	3	2	3	First	No	No	No	No
40	M005022	Muscogee	3	SR 85 FROM SR 1 TO SR 1	Resurface & Maintenance	3	3	3	First	No	No	No	No
41	M006023	Dade	6	I-59 FROM SR 136 TO I-24	Pavement Rehab	3	3	3	First	No	No	Yes	No
42	M006052	Camden	5	I-95 FM S OF HORSE STAMP CHURCH ROAD TO LITTLE SATILLA RIVER	Pavement Rehab	3	2	3	First	No	Yes	Yes	Yes
43	M006228	Bibb, Monroe	3	I-75 FROM CR 85/ARKWRIGHT ROAD/BIBB TO I-475/MONROE	Resurface & Maintenance	3	3	3	First	No	No	No	No
44	M006260	Chattooga	6	SR 1/US 27 FROM CHATTOOGA RIVER TO CHATTOOGA RIVER	Resurface & Maintenance	3	3	3	First	No	No	Yes	No
45	M006402	Burke	2	SR 24 FROM SR 121 BYPASS TO SR 56	Resurface & Maintenance	3	2	3	First	No	No	No	No
46	M006430	Chattooga	6	SR 1/US 27 FROM CR 329/GORE SUBLIGNA ROAD TO CHATTOOGA RIVER	Resurface & Maintenance	3	3	3	First	No	No	Yes	No
47	M006446	DeKalb	7	I-285 FROM CR 5158/HENDERSON MILL ROAD TO SR 8	Concrete Rehab	3	2	3	First	No	No	Yes	Yes

No.	Project ID	County	GDOT District	Description	Primary Work Type	Criticality Class	Vulnerability Class	Risk Class	Priority	On Evacuation Route	Coastal County	1% Annual Chance Flood Hazard Area	0.2% Annual Chance Flood Hazard Area
48	M006462	DeKalb	7	SR 155 FROM I-20 TO SR 10	Resurface & Maintenance	2	3	3	First	No	No	Yes	Yes
49	M006464	Fulton	7	SR 13 FROM SR 9 TO NORTH FORK PEACHTREE CREEK	Resurface & Maintenance	2	3	3	First	No	No	Yes	Yes
50	M006466	Douglas	7	SR 5 FROM CARROLL COUNTY LINE TO N OF CR 188/FLINT HILL ROAD	Resurface & Maintenance	2	3	3	First	No	No	No	No
51	M006467	Clayton	7	SR 138 SPUR FROM SR 138 TO SR 3	Resurface & Maintenance	2	3	3	First	No	No	No	No
52	M006472	Worth	4	SR 33 FROM COLQUITT COUNTY LINE TO CS 719/CROCKETT STREET	Resurface & Maintenance	2	3	3	First	No	No	Yes	No
53	M006477	Floyd	6	SR 156 FROM SR 1 TO GORDON COUNTY LINE	Resurface & Maintenance	1	3	3	Second	No	No	Yes	No
54	M006489	Clayton, Fulton	7	I-85 @ I-285 IN CLAYTON & FULTON COUNTY; INC 22 RAMPS	Resurface & Maintenance	3	2	3	First	No	No	No	No
55	M006499	Henry	3	SR 20 FROM INDUSTRIAL BLVD TO LOWER WOOLSEY ROAD	Resurface & Maintenance	2	3	3	First	No	No	No	No
56	M006524	DeKalb	7	SR 154 FROM FULTON COUNTY LINE TO SR 10	Resurface & Maintenance	2	3	3	First	No	No	Yes	Yes
57	M006536	Gwinnett	1	SR 378 FROM SR 13 TO SR 8	Resurface & Maintenance	2	3	3	First	No	No	Yes	No
58	M006537	Gwinnett, Walton	1	SR 20 FROM SR 124/GWINNETT TO COVINGTON STREET/WALTON	Resurface & Maintenance	1	3	3	Second	No	No	No	No
59	M006539	Jackson	1	SR 98 FROM WASHINGTON STREET TO SR 334	Resurface & Maintenance	2	3	3	First	No	No	No	No
60	M006541	Richmond	2	SR 4/SR 540/US 1 FM SOUTH PRONG CREEK TO N OF BRENTWAY DRIVE	Resurface & Maintenance	3	3	3	First	No	No	Yes	Yes
61	M006545	Decatur	4	SR 38/US 84 FROM SR 38BU TO GRADY COUNTY LINE	Resurface & Maintenance	3	2	3	First	No	No	Yes	Yes
62	M006549	Liberty	5	SR 38/US 84 FROM W OF SR 119 TO E OF WALLACE MARTIN DRIVE	Resurface & Maintenance	3	2	3	First	Yes	Yes	No	Yes
63	M006552	Murray	6	SR 61/US 411 FROM S OF RHYMER ROAD TO SR 2	Resurface & Maintenance	2	3	3	First	No	No	Yes	No
64	0008458	Cook	4	I-75 @ CR 246/KINARD BRIDGE ROAD - PHASE II	Interchange	3	1	2	Third	Yes	No	Yes	Yes
65	0010298	Lowndes	4	I-75 @ SR 133 - PHASE II	Interchange	3	1	2	Third	Yes	No	No	No
66	0013156	Cook	4	I-75 FROM LOWNDES COUNTY LINE TO SR 37 - PHASE II	Interchange	3	1	2	Third	Yes	No	Yes	Yes
67	0013157	Cook	4	I-75 @ CR 251/ROUNTREE BR RD & @ CR 253/BARNEYVILLE RD-PH II	Interchange	3	1	2	Third	Yes	No	No	No

No.	Project ID	County	GDOT District	Description	Primary Work Type	Criticality Class	Vulnerability Class	Risk Class	Priority	On Evacuation Route	Coastal County	1% Annual Chance Flood Hazard Area	0.2% Annual Chance Flood Hazard Area
68	0013677	Clinch	5	SR 89/US 441 FM CR 40/ANTIOCH CHURCH RD TO CR 549/WHEELER ST	Widening	3	2	2	First	Yes	No	Yes	No
69	0013808	Franklin	1	SR 106 @ NAILS CREEK 7 MI S OF CARNESVILLE	Bridges	3	2	2	First	No	No	No	No
70	0013856	Crisp	4	I-75 @ SR 33 CONN	Interchange	3	1	2	Third	Yes	No	No	No
71	0017188	Tattnall	5	SR 23/SR 73 DRAINAGE IMPROVEMENTS IN GLENNVILLE	Drainage Improvements	2	2	2	First	Yes	No	No	No
72	0017221	Bibb, Houston	3	SR 247 @ ECHECONNIE CREEK & OVERFLOW @ 3 LOCS	Bridges	3	2	2	First	No	No	Yes	No
73	0017271	Chatham	5	I-95 @ SR 21	Interchange	3	2	2	First	Yes	No	Yes	Yes
74	0017443	Glynn	5	SR 32/SR 99 FROM SR 99 TO SR 27/US 25/US 341	Widening	2	2	2	First	Yes	Yes	Yes	Yes
75	0019083	Franklin	1	I-85 SB & NB @ NORTH FORK BROAD RIVER 5 MI SW OF LAVONIA	Bridges	3	2	2	First	No	No	Yes	No
76	0019103	Pickens	6	SR 136 @ CR 594/BIG OAK DRIVE	Culvert	1	3	2	Second	No	No	No	No
77	0019294	Lumpkin	1	SR 60/US 19 @ AUSTIN WAY	Slope Repair	2	2	2	First	No	No	No	No
78	0019355	Lumpkin	1	SR 60 FROM N OF BEAR PAW RIDGE TO S OF YAHOOOLA CREEK	Slope Repair	2	2	2	First	No	No	Yes	Yes
79	0019472	Thomas	4	SR 3/SR 300 @ 4 LOCS IN THOMAS COUNTY - VRU	Intersection Improvement	3	2	2	First	Yes	No	No	No
80	0020040	DeKalb	7	SR 8/US 23 FROM SOUTH PONCE DE LEON AVE TO RIDGECREST ROAD	Drainage Improvements	3	2	2	First	No	No	No	No
81	0020168	Bryan, Bulloch, Chatham, Effingham	5	I-16 FROM I-95/CHATHAM TO SR 67/BULLOCH	Widening	3	2	2	First	Yes	Yes	Yes	No
82	0020172	Chatham, Effingham	5	SR 21 FROM SR 30/CHATHAM TO 9TH ST/EFFINGHAM	Widening	2	2	2	First	Yes	Yes	Yes	Yes
83	M005881	Clarke, Oconee	1	SR 10/US 78 FROM WALTON COUNTY LINE TO SR 8	Resurface & Maintenance	2	2	2	First	No	No	No	No
84	M006422	Wayne	5	SR 38/US 84 FROM W OF CS 1041/SUNSET BLVD TO S OF SR 23	Resurface & Maintenance	3	2	2	First	No	No	Yes	No
85	M006503	Jones	3	SR 22 FROM GRAY BYPASS TO BALDWIN COUNTY LINE	Resurface & Maintenance	2	2	2	First	No	No	Yes	No
86	M006528	Early	4	SR 216 FROM BAKER COUNTY LINE TO SR 45	Resurface & Maintenance	1	2	2	Second	No	No	No	No
87	M006694	Lowndes	4	I-75 SB & NB @ WEIGH STATION IN LOWNDES COUNTY	Resurface & Maintenance	3	1	2	Third	Yes	No	No	No
88	0018336	Sumter	3	SR 377 @ CS 789/CHURCH STREET	Drainage Improvements	3	1	1	Third	No	No	No	No
89	0019099	Terrell	4	SR 55 @ CLARKS MILLPOND DAM	Culvert	1	1	1	Last	No	No	Yes	No

No.	Project ID	County	GDOT District	Description	Primary Work Type	Criticality Class	Vulnerability Class	Risk Class	Priority	On Evacuation Route	Coastal County	1% Annual Chance Flood Hazard Area	0.2% Annual Chance Flood Hazard Area
90	0019100	Fannin	6	SR 60 @ TOCCOA RIVER	Slope Repair	1	1	1	Last	No	No	Yes	No
91	0019104	Pickens	6	SR 136 @ 0.40 MI NW OF CR 198/SWAN BRIDGE ROAD	Culvert	1	2	1	Second	No	No	Yes	No
92	0019126	Colquitt	4	CR 480/LOWER MEIGS ROAD @ BRIDGE CRK & @ BRIDGE CRK OVERFLOW	Bridges	2	1	1	Third	No	No	Yes	No
93	0020297	Bulloch	5	SR 26 @ SR 119	Turn Lanes	2	1	1	Third	Yes	No	No	No
94	M006273	McDuffie	2	I-20 FROM SR 10/US 78 TO 0.17 MI W OF GREENBRIAR CREEK	Concrete Rehab	3	1	1	Third	No	No	Yes	No
95	M006274	Columbia, McDuffie	2	I-20 FM 0.17 MI W OF GREENBRIAR CREEK TO 0.40 MI W OF SR 47	Concrete Rehab	3	1	1	Third	No	No	Yes	Yes
96	M006417	Montgomery	5	SR 56 FROM TOOMBS COUNTY LINE TO TREUTLEN COUNTY LINE	Resurface & Maintenance	2	1	1	Third	No	No	Yes	No
97	M006445	Greene	2	I-20 FROM W OF SR 44 TO SR 15	Concrete Rehab	3	1	1	Third	No	No	Yes	No
98	M006548	Appling, Wayne	5	SR 169 FROM SUNSET BLVD/WAYNE TO S OF SR 121/APPLING	Resurface & Maintenance	1	2	1	Second	No	No	Yes	No
99	M006584	Jones	3	SR 11 FROM SR 22 TO JASPER COUNTY LINE	Resurface & Maintenance	1	2	1	Second	No	No	No	No
100	0019102	Lumpkin	1	GARWOOD LAKE DAM DECOMMISSION & REMOVAL	Drainage Improvements	0	0	0	First	No	No	No	No
101	0019372	Catoosa	6	CITY OF FORT OGLETHORPE RESURFACING @ 6 LOCS	Resurface & Maintenance	0	0	0	First	No	No	No	No
102	0019453	Catoosa, Walker	6	CS 550/THOMAS ROAD & CS 647/FANT DRIVE IN FORT OGLETHORPE	Resurface & Maintenance	0	0	0	First	No	No	Yes	Yes
103	0019928	Laurens	2	OFF-SYSTEM SAFETY IMPROVEMENTS @ 1 LOC IN LAURENS CO - HRRR	Shoulder Work	0	0	0	First	No	No	No	No
104	0019929	Oglethorpe	2	OFF-SYSTEM SAFETY IMPROVEMENTS @ 5 LOC IN OGLETHORPE CO-HRRR	Shoulder Work	0	0	0	First	No	No	Yes	No
105	0019951	Schley	3	OFF-SYSTEM SAFETY IMPROVEMENTS @ 1 LOC IN SCHLEY COUNTY-HRRR	Shoulder Work	0	0	0	First	No	No	No	No
106	0019960	Walton	1	OFF-SYSTEM SAFETY IMPROVEMENTS @ 1 LOC IN WALTON COUNTY	Shoulder Work	0	0	0	First	No	No	No	No
107	M006664	Colquitt	4	MILL POND ROAD FM FUNSTON SALE CITY ROAD TO OLD DOERUN ROAD	Resurface & Maintenance	0	0	0	First	No	No	Yes	No

Table 6. Projects Approved for PROTECT funding by FHWA as of October 2024

No.	Project ID	Description	Primary Work Type	County	GDOT District	Eligibility Type	Asset Type	Approved Amount	Federal Cost Share
1	0007055	SR 180 @ SLAUGHTER CREEK	Bridges	Union	1	Resilience Improvement	Bridge	\$1,084,329.00	Incremental
2	0015541	SR 58 @ SQUIRREL TOWN CREEK 4 MI NE OF TRENTON	Bridges	Dade	6	Resilience Improvement	Bridge	\$1,655,706.00	Incremental
3	0013717	SR 54 @ BEAR CREEK 3.5 MI SE OF MORELAND	Bridges	Coweta	3	Resilience Improvement	Bridge	\$1,390,196.00	Incremental
4	0015557	SR 8/SR 174 @ HUDSON RIVER 8.8 MI N OF DANIELSVILLE	Bridges	Franklin	1	Resilience Improvement	Bridge	\$1,009,224.65	Incremental
5	0015621	CR 454/LAMBS BRIDGE ROAD @ CANOOCHEE CREEK NE OF SWAINSBORO	Bridges	Emanuel	2	Resilience Improvement	Bridge	\$984,000.00	Incremental
6	0015540	SR 7/US 41 @ CEDAR CREEK 1.5 MI S OF CORDELE	Bridges	Crisp	4	Resilience Improvement	Bridge	\$783,841.86	Incremental
7	0015611	VANILLA LANE @ BIG TURNIPTOWN CREEK 4.9 MI NE OF ELLIJAY	Bridges	Gilmer	6	Resilience Improvement	Bridge	\$300,221.24	Incremental
8	0015622	CR 454/LAMBS BRIDGE RD@LITTLE CANOOCHEE CRK NE OF SWAINSBORO	Bridges	Emanuel	2	Resilience Improvement	Bridge	\$1,572,246.84	Incremental
9	M006391	SR 40 FROM W OF CS 599/KINGS BAY ROAD TO E OF SR 40 SPUR	Resurface & Maintenance	Camden	5	Community Resilience & Evacuation Route	Roadway	\$2,641,112.63	Full
10	0019359	SR 144 SPUR @ OGEECHEE RIVER	Bridges	Bryan	5	At-Risk Coastal Infrastructure	Bridge	\$10,075,000.00	Full
11	0013719	SR 38/US 84 @ DOCTORS CREEK 3 MI E OF LUDOWICI	Bridges	Long	5	Resilience Improvement	Bridge	\$1,647,848.00	Incremental
12	0013738	SR 25 @ WHITEOAK CREEK 5 MI N OF WOODBINE	Bridges	Camden	5	Resilience Improvement	Bridge	\$786,660.00	Incremental
13	0015601	CR 133/EBENEZER ROAD @ MUCKALEE CREEK 7.5 MI W OF ELLAVILLE	Bridges	Schley	3	Resilience Improvement	Bridge	\$987,874.00	Incremental
14	0015615	CR 422/JEFF DAVIS ROAD @ TEN MILE CREEK 2 MI NW OF THOMASTON	Bridges	Upson	3	Resilience Improvement	Bridge	\$800,000.00	Incremental
15	0014072	I-16 EB & WB @ WALNUT CREEK 1 MI E OF MACON	Bridges	Bibb	3	Resilience Improvement; Community Resilience and Evacuation Route Project	Bridge	\$17,017,539.00	Full
16	0015641	CR 305/TYRE BRIDGE RD @ SIXTY FOOT CREEK N OF PATTERSON CITY	Bridges	Pierce	5	Resilience Improvement	Bridge	\$448,000.00	Incremental

Next Steps

The RIP and the associated eligible project lists are dynamic and will be updated periodically as new data becomes available.

GDOT will continue to utilize its Research Program to partner with researchers from the Georgia Transportation Institute (GTI) to better understand the changing climate hazards, their impacts on the transportation system, and to prepare for resilience improvement based on asset vulnerability, criticality, and risk. Additionally, the Resiliency Committee will actively engage all offices and functions within GDOT to incorporate resilience in project identification, planning, and design. This process should involve the participation from the MPOs, partners at the regional and federal levels, as well as local stakeholders to ensure the resilience improvement efforts are customized to local needs.

APPENDIX A

Plan Review

The table below lists the plans/tools that were reviewed prior to the development of the GDOT RIP to note best practices.

Plan Peer Review

State	Agency	Plan/Tool	Plan Date
Alaska	DCCED	Alaska Risk Mapping, Assessment and Planning (Risk MAP) Program	2023
Arizona	ADOT	Asset Management, Extreme Weather, and Proxy Indicators Pilot Project	2020
Arkansas	ARDOT	Electric Vehicle Infrastructure Deployment Plan	2022
California	CalSTA	Climate Action Plan for Transportation Infrastructure (CAPTI)	2021
California	Caltrans	Climate Change Vulnerability Assessment	2021
Colorado	CDOT	Risk and Resiliency Analysis Risk and Resiliency Excel Tool	2020
Connecticut	CTDOT	Connecticut's Charging Ahead Plan	2022
Florida	FDOT	Resilience Action Plan	2023
Hawaii	HDOT	Climate Adaption Action Plan Climate Adaption Action Plan Exposure Assessment	2021
Illinois	IDOT	All Hazard Transportation System Vulnerability Assessment	2017
Kansas	KDOT	KDOT Resilience Improvement Plan (RIP)	Under development
Louisiana	DOTD	Louisiana State Plan for Electric Vehicle Infrastructure Deployment	2022
Minnesota	MnDOT	Transportation Resilience: Current Practices and Opportunities for MnDOT	2020
Minnesota	MnDOT	MnDOT Resilience Improvement Plan (RIP)	Under development
North Carolina	NCDOT	NCDOT 2022 Resilience Strategy Report	2022
North Carolina	NCDEQ	Climate Risk Assessment and Resilience Plan	2020
Ohio	ODOT	ODOT Infrastructure Resiliency Plan	2016
Ohio	ODOT	Access Ohio 2045 System Resiliency White Paper	2020
Oregon	OEM	Oregon Resilience Plan	2018
Oregon	Dept of Energy	Oregon Guidebook for Local Energy Resilience	2019
Pennsylvania	PennDOT	Extreme Weather Vulnerability Study	2017
Rhode Island	State	Resilient Rhody	2018
Tennessee	TDOT	Assessing the Vulnerability of Tennessee Transportation Assets to Extreme Weather	2015
Texas	TxDOT	Statewide Resiliency Plan	Under development

State	Agency	Plan/Tool	Plan Date
Utah	UDOT	UDOT Asset Risk Management Process Risk Priority Analysis Story Map	2020
Vermont	VTRANS	NEV Infrastructure Plan	2020
Vermont	VTRANS	Transportation Resilience Planning Tool	2023
Virginia	VDOT	VDOT Resilience Plan	2022
Washington	WSDOT	Climate Impacts Vulnerability Assessment	2011
Wyoming	WYDOT	Risk and Resiliency Plan for Critical Tourism Transportation Assets Risk and Resiliency Plan for Critical Freight Transportation Assets	2018
Wyoming	WYDOT	NEVI Deployment Plan	2022
Wyoming	WYDOT	State of Wyoming Transportation Risk and Resilience Improvement Plan	Under development

The table provides an inventory of which plans included a vision statement, goals, objectives, and/or policies. This information was considered before the establishment of the GDOT RIP goals.

Inclusion of Vision, Goals, Objectives, and/or Policies in Peer Plans

State	Vision/Intent	Goals/Principles	Objectives/Strategies	Policies/Actions
Arizona	-	X	-	-
California	X	X	X	X
Hawaii	X	-	X	-
Minnesota	-	X	-	-
North Carolina	-	X	X	X
Ohio	-	-	X	X
Pennsylvania	-	-	X	X
Tennessee	-	-	X	X
Utah	-	X	X	-
Virginia	-	X	X	X
Wyoming	-	X	X	X

APPENDIX B

FHWA PROTECT Eligibility Screening Form

FHWA GA Division PROTECT Eligibility Screening Form

STIP/TIP Project Information					
Project Name:					
Project Number:					
STIP/TIP Number:					
GDOT District:					
Planning Reference Document	Not Currently in Planning Document				
Applicable STIP/TIP Project Cost(s):	PE	ROW	UTL	CST	Total
					\$ 0
Project Description:					
Project Location Map (click link to upload):					

PROTECT Eligibility	
Based on the <i>PROTECT Formula Program Implementation Guidance</i> , GDOT has determined that the project is eligible for PROTECT Formula Program funds as a(n):	
☐	Resilience Improvement Project
☐	Community Resilience and Evacuation Route Project
☐	At-Risk Coastal Infrastructure Project
Please include justification for this determination in the box below:	

Federal Cost Share			
Please select one of the following options:			
☐	OPTION 1: PROTECT Formula Program funds will be used to cover the incremental cost of making the infrastructure more resilient.		
☐	OPTION 2: The primary purpose of the project is to address resilience. Therefore, GDOT is requesting that the project be fully funded (up to the applicable Federal share) using PROTECT Formula Program funds.		
If Option 1 is selected above, please provide specific information on design elements that would make the infrastructure more resilient. In addition, please provide supporting documentation showing how the incremental cost percentage was calculated. Requests for PROTECT Formula Program funds in FMIS shall not exceed the incremental cost percentage identified below.			
Asset Type	Resilience Improvement	Incremental Cost Percentage	
Bridge		PE	0%
		ROW	0%
		UTL	0%
		CST	0%

If Option 2 is selected above, please provide supporting documentation which demonstrates that the primary purpose of the project is to address resilience.	
☐ Yes	Is the project eligible for an authorized reduction in the non-Federal share?
☐ No	
If yes, please select all that apply from the options below:	
☐	The project is eligible for a 7% reduction in the non-Federal share because the State has developed a Resilience Improvement Plan (RIP) and prioritized the project on that RIP.
☐	The project is eligible for a 3% reduction in the non-Federal share because the RIP is incorporated into the metropolitan transportation plan under 23 U.S.C. 134 or the statewide long-range transportation plan under 23 U.S.C. 135, as applicable.

Approvals		
Based on the information provided in this form, and the associated supporting documentation, GDOT has determined that the project is eligible for PROTECT Formula Program funds. An estimate of the requested PROTECT Formula Program funds is provided below.		
	PE:	
	ROW:	
	UTL:	
	CST:	
	Total:	\$ 0
GDOT Approver		
FHWA Approver		